Contract enforcement and international trade: a new look

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This paper reexamines the role of contract-enforcement institutions in international trade. The standard view is that countries with stronger domestic institutions specialize in contract-intensive sectors because weak enforcement discourages relationship-specific investment. I argue that this view is incomplete in an open economy: firms may source specific inputs from abroad, and contracting frictions may depend on the institutions of both the input supplier and the input buyer. I develop a multi-country model in which incomplete contracts generate an industry-specific bilateral cost wedge for relationship-specific inputs. The model distinguishes two channels. First, contracting frictions raise the effective trade cost of specific inputs. Second, they raise production costs in sectors that rely heavily on those inputs. The model also implies that exporter and importer institutions are substitutes, and that access to foreign suppliers may partly mitigate weak domestic institutions. Using bilateral product-level trade data and measures of rule of law, I find evidence for both cost channels. Trade in specific inputs is increasing in the quality of contract enforcement in both origin and destination countries, and the marginal effect of one country's institutions is weaker when the other country's institutions are stronger. Domestic institutions also matter more in sectors that rely heavily on domestic specific inputs and in industries where know-how transfers are important. However, the direct evidence that access to foreign institutions offsets weak domestic institutions is limited. The results suggest that domestic institutions remain central to comparative advantage, but their effects operate through global input sourcing as well as domestic production.

1. Introduction

Contract enforcement institutions are an important determinant of comparative advantage.[1] The standard explanation is that weak enforcement discourages relationship-specific investment and thereby raises production costs in sectors where these investments are important. Empirically, this dependence has been measured in several related but distinct ways: as the complexity of the production process (Levchenko 2007), as the differentiation or complexity of traded goods (Berkowitz, Moenius, and Pistor 2004, 2006), and as the share of intermediate inputs requiring relationship-specific investments (Nunn 2007).[2]

We argue that this explanation is incomplete for two reasons. First, existing theories focus primarily on the export side of international trade and largely abstract from the import side, implicitly assuming that specific inputs must be sourced domestically (Nunn and Trefler 2014). If specific inputs can instead be imported, firms in countries with weak domestic institutions may partially circumvent domestic contracting frictions by sourcing these inputs from countries with stronger institutions. Second, existing theories give limited attention to the role of importer institutions in shaping trade in contract-intensive goods. Although some studies emphasize institutions in the importing country (Anderson and Marcouiller 2002; Berkowitz, Moenius, and Pistor 2006; Schmidt-Eisenlohr 2013), they focus on channels that apply broadly across goods, such as bribe costs, expropriation risk, or the risk of non-payment. We provide a rationale for, and show empirically, that importer institutions matter especially for trade in specific inputs.

This paper addresses both gaps. We extend the contracting-frictions model of Acemoglu, Antràs, and Helpman (2007) in two directions. First, we relax their symmetry assumption across input suppliers, allowing suppliers to differ in the institutional environment, production costs, and trade costs associated with their location. This extension is necessary to accommodate international sourcing, where inputs may be supplied from

[1] Potentially even more important than factor endowments: Nunn (2007) finds that the partial effect of judicial quality on trade patterns exceeds the combined effects of capital and skilled labor.

[2] Berkowitz, Moenius, and Pistor (2004, 2006) and Nunn (2007) construct their measures using the classification developed by Rauch (1999), but they use it to capture different objects. Rauch (1999) groups internationally traded commodities into three categories: goods traded on organized exchanges, goods that are not exchange-traded but for which reference prices are available, and goods that are neither exchange-traded nor reference-priced. Berkowitz, Moenius, and Pistor (2004, 2006) use this classification *directly* as a measure of product complexity, interpreting exchange-traded goods as simple and non-exchange-traded goods, especially those that are neither exchange-traded nor reference-priced, as more complex. Nunn (2007), by contrast, uses Rauch's classification *indirectly*: he applies it to upstream intermediate inputs, interprets market thickness as inversely related to relationship-specificity, and then classifies downstream final goods as more contract-intensive when they rely more heavily on relationship-specific inputs. Levchenko (2007) also classifies sectors based on their input structure, but does not rely on Rauch's classification. Instead, he uses the Herfindahl index of intermediate input use as a proxy for a sector's institutional dependence or complexity. In this paper, we refer to intermediate inputs that are neither exchange-traded nor reference-priced as *specific inputs* and to Nunn's input-weighted downstream measure as a sector's *contract intensity*.

different countries (Antràs, Fort, and Tintelnot 2017). Second, we allow both buyers and suppliers to make relationship-specific investments, as in Antràs and Helpman (2004) and Antràs and Helpman (2008). Despite these extensions, the model remains tractable: contracting frictions can be summarized by an industry-specific bilateral wedge that operates much like an iceberg trade cost. This representation makes it straightforward to embed contracting frictions into standard multi-country trade models with minimal modification. We do so in a multi-sector model with input-output linkages, following Caliendo and Parro (2015).

The enriched model yields five main insights. First, contracting frictions operate as cost distortions that raise prices. Second, stronger contract-enforcement institutions mitigate these distortions. Third, firms in countries with weak domestic institutions can partially offset these costs by importing specific inputs from countries with stronger institutional environments. Fourth, importer and exporter institutions are *substitutes*: stronger institutions in one country reduce the marginal benefit of stronger institutions in the other. Firms in weak institutional environments can therefore rely partly on their trading partners' institutions, suggesting that existing empirical work may simultaneously overstate the importance of domestic institutional quality and understate the importance of institutional quality more broadly.

Fifth, the model identifies two main channels through which domestic institutions continue to shape international trade patterns, even when specific inputs can be traded. First, strong domestic institutions remain important to export specific inputs themselves, as in Berkowitz, Moenius, and Pistor (2004, 2006). We call this the *trade cost channel*. Second, weak domestic institutions are harder to circumvent in industries where production relies heavily on relationship-specific investments by input buyers (interpreted as *know-how transfers*), rather than only by suppliers (interpreted as *assembly services*), or when specific inputs are non-tradable. We call this the *production channel*.

The empirical analysis tests these predictions using bilateral product-level trade data for 2012, 2015, and 2018. Products are classified as final goods, generic inputs, or specific inputs, and the regressions exploit variation across product types within detailed origin–destination–industry–year cells. The results support the two main cost channels. Consistent with the trade cost channel, trade in specific inputs is increasing in the quality of contract enforcement in both the origin and the destination country. Moreover, the interaction between origin and destination institutions is negative, consistent with the model's prediction that institutions in the two countries are substitutes. Consistent with the production cost channel, domestic contract enforcement matters more for exports in countries and industries that rely more heavily on domestic suppliers of specific inputs. The evidence also indicates that domestic institutions matter especially in contract-intensive industries where know-how transfers are important. However, the direct evidence that access to foreign institutions allows countries to circumvent weak domestic institutions is more limited: exposure to foreign suppliers with stronger institutions is positively associated with exports, but the estimates are not statistically significant.

This paper contributes first to the literature on institutions and comparative advantage. Levchenko (2007), Nunn (2007), Berkowitz, Moenius, and Pistor (2004, 2006), and Costinot (2009) show that countries with stronger institutions specialize in goods or sectors that are more institutionally dependent. This paper preserves the central insight that contract enforcement is a source of comparative advantage, but changes the object of analysis. Rather than treating institutional quality as a country-level characteristic that affects production costs through domestic institutions alone, the model shows that contracting frictions are bilateral, input-specific, and mediated by sourcing decisions. This distinction matters empirically because the institutional content of trade depends not only on what a country produces, but also on where its specific inputs are produced and which country's contract enforcement institutions discipline the relevant relationship-specific investments.

Second, the paper contributes to the literature on incomplete contracts and the organization of international production. A large body of work studies how contractual frictions shape outsourcing, integration, technology adoption, multinational production, and global value chains (Grossman and Hart 1986; Hart and Moore 1990;Antràs 2003, 2005; Antràs and Helpman 2004, 2008; Grossman and Helpman 2002;Acemoglu, Antràs, and Helpman 2007; Antràs and Chor 2013). Relative to this literature, the paper focuses less on firm boundaries and more on the trade-flow implications of contracting frictions when inputs are sourced across countries. The model implies that incomplete contracts distort not only the organizational form of production but also the effective bilateral cost of specific inputs. This generates predictions about the composition of trade across final goods, generic inputs, and specific inputs that can be taken to the data using standard gravity methods.

Third, the paper connects the institutional-comparative-advantage literature to work on production sharing and input-output linkages in trade (Johnson and Noguera 2012; Caliendo and Parro 2015; Antràs, Fort, and Tintelnot 2017; Antràs et al. 2024). In models with global sourcing, imported inputs can influence downstream comparative advantage. This paper shows that once specific inputs can be traded, the effect of institutions on trade must be decomposed into a trade cost channel, which affects bilateral flows of specific inputs, and a production cost channel, which affects the cost of downstream production through the price index of specific inputs. This decomposition helps reconcile why domestic institutions remain empirically important even when firms have access to foreign suppliers.

This paper is organized as follows. Section 2 motivates the paper by showing that access to foreign institutions affects countries' export patterns even after controlling for domestic institutions. Sections 3 and 4 present the model. Section 3 develops the input-sourcing problem under incomplete contracts, and section 4 embeds the resulting cost wedge into a multi-country trade model with input-output linkages. Section 5 describes the data and variable construction. Section 6 presents the empirical strategy and results. Section 7 concludes.

2. Motivation

A first limitation of existing theories linking domestic institutions to comparative advantage is that they focus on the export side of international trade while largely abstracting from the import side. In doing so, they implicitly assume that specific inputs must be sourced domestically. If specific inputs can instead be imported, firms in countries with weak domestic institutions may partially circumvent those weaknesses by sourcing contract-intensive inputs from abroad.

Mexico illustrates this possibility. Figure 1 reproduces the central empirical relationship in Nunn (2007): countries with stronger contract enforcement institutions tend to specialize in contract-intensive sectors, as shown by the gray points.[3] The figure relates *expected* comparative advantage in contract-intensive sectors—measured by the interaction between a country's quality of domestic contract enforcement, Q_c, and an industry's contract intensity, z_i—to *revealed* comparative advantage, measured by industry-country export flows, x_{ic}. The relationship is estimated after controlling for industry fixed effects, α_i, country fixed effects, α_c, and two standard factor-proportion determinants of comparative advantage: the interactions between countries' endowments of skilled labor and capital, H_c and K_c, and industries' skill and capital intensities, h_i and k_i. In other words, the figure isolates the variation identifying β_1 in:[4]

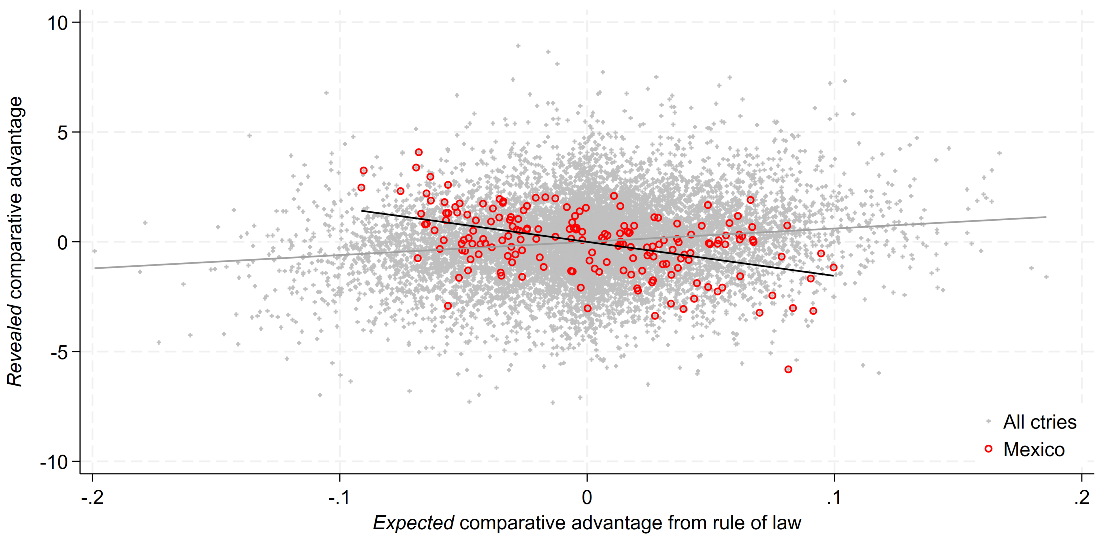
$$\log x_{ic} = \alpha_i + \alpha_c + \beta_1 \left(z_i \times Q_c\right) + \beta_2 \left(h_i \times H_c\right) + \beta_3 \left(k_i \times K_c\right) + \varepsilon_{ic} \tag{1}$$

[3]The figure was constructed using the original data from Nunn (2007).

[4]By the Frisch-Waugh-Lovell theorem, $\hat{\beta}_1$ is the coefficient from a regression of the component of $\log x_{ic}$ orthogonal to the fixed effects and the two factor-proportion interactions on the component of $z_i \times Q_c$ orthogonal to the same controls. Figure 1 plots these residualized variables, with each point corresponding to an industry-country pair. The y-axis, labeled "*Revealed* comparative advantage," captures the residual variation in exports after removing industry and country averages and the two factor-proportion controls. The x-axis, labeled "*Expected* comparative advantage from rule of law," captures the residual variation in the institutions-based comparative advantage term after applying the same controls. Before residualizing with respect to the factor-proportion interactions, the two-way demeaned interaction can be written as $\left(\widetilde{z_i \times Q_c}\right) = \left(z_i - \bar{z}\right)\left(Q_c - \bar{Q}\right)$, while the two-way demeaned dependent variable is $\widetilde{\log x_{ic}} \equiv \log x_{ic} - \overline{\log x_{i\cdot}} - \overline{\log x_{\cdot c}} + \overline{\log x_{\cdot\cdot}}$. This residualized export measure can be interpreted as *revealed comparative advantage* because a positive value indicates that a country exports more in an industry than would be predicted by the country's average exports across industries and the industry's average exports across countries. Similarly, the residualized interaction captures *expected comparative advantage* from contract enforcement: industry-country pairs receive high values when contract intensity and institutional quality are aligned in the direction predicted by institutional comparative advantage.

Mexico, highlighted in red, appears to defy the general pattern. Despite below-average domestic contract-enforcement institutions, Mexico specializes in relatively contract-intensive sectors. One interpretation is that Mexican firms are not constrained only by domestic institutions: they may also draw on stronger foreign institutional environments by sourcing specific inputs from abroad.

FIGURE 1. Contract enforcement and comparative advantage



Note: The figure plots the partial relationship between log exports and comparative advantage in contract-intensive industries, controlling for industry and country fixed effects and for interactions between factor endowments and factor intensities. "Rule of law" is from Kaufmann, Kraay, and Mastruzzi (2004), and "contract intensity" is from Nunn (2007). Mexico is highlighted in red. The y-axis measures export specialization: higher values indicate that a country exports more in a given industry relative to both its average exports across industries and the world's average exports in that industry. The x-axis measures the component of expected comparative advantage associated with the interaction between industry contract intensity and domestic rule of law, net of the same controls. A positive slope indicates that countries with stronger contract enforcement specialize in more contract-intensive industries.

We can assess the empirical relevance of this hypothesis by extending equation 1 to include a measure of access to foreign institutions. Let $Q_{c'}$ denote institutional quality in foreign country c', and let $d_{cc'}$ denote bilateral distance between countries c and c'. We construct a reduced-form index of country c's exposure to foreign institutional quality as a distance-weighted average of institutional quality abroad:

$$FQ_c = \sum_{c' \neq c} w_{cc'} Q_{c'}, \qquad \text{where:} \qquad w_{cc'} = \frac{d_{cc'}^{-1}}{\sum_{l \neq c} d_{cl}^{-1}} \tag{2}$$

where the weights are normalized inverse-distance weights, so nearby countries receive more weight and the weights sum to one for each country c.

Adding the interaction $z_i \times FQ_c$ to the regression reduces the magnitude of the coefficient on domestic institutional comparative advantage, $z_i \times Q_c$, but the coefficient remains statistically significant. This confirms the robustness of the central insight in

Nunn (2007): domestic contract enforcement remains an important source of comparative advantage. At the same time, the coefficient on $z_i \times FQ_c$ is statistically significant and quantitatively similar, suggesting that access to foreign institutions also shapes specialization in contract-intensive sectors.

Figure 2 illustrates this result. In the augmented regression, Mexico's export specialization remains difficult to reconcile with expected comparative advantage based only on domestic institutions. However, it is consistent with expected comparative advantage based on access to foreign institutions. This pattern is consistent with the hypothesis that firms in countries with weak domestic institutions can partially overcome domestic contracting frictions by sourcing specific inputs from countries with stronger institutional environments.

3. A model of input sourcing under incomplete contracts

We present the model in two parts. The first part studies firms' input-sourcing decisions. Firms choose suppliers from around the world subject to contracting frictions. Our model extends the framework of Acemoglu, Antràs, and Helpman (2007) in two ways. First, we relax the symmetry assumption and allow input suppliers to differ across countries. Second, we allow input quality to depend not only on suppliers' choices but also on buyers' choices, creating a role for the institutions of the importing country, as in Antràs and Helpman (2004) and Antràs and Helpman (2008). We show that the effect of contracting frictions can be summarized by a cost wedge with the properties of an iceberg trade cost. This makes it straightforward to incorporate contracting frictions into otherwise standard models of international trade. The second part uses this result to embed contracting frictions in a multi-sector trade model, similar to Caliendo and Parro (2015).

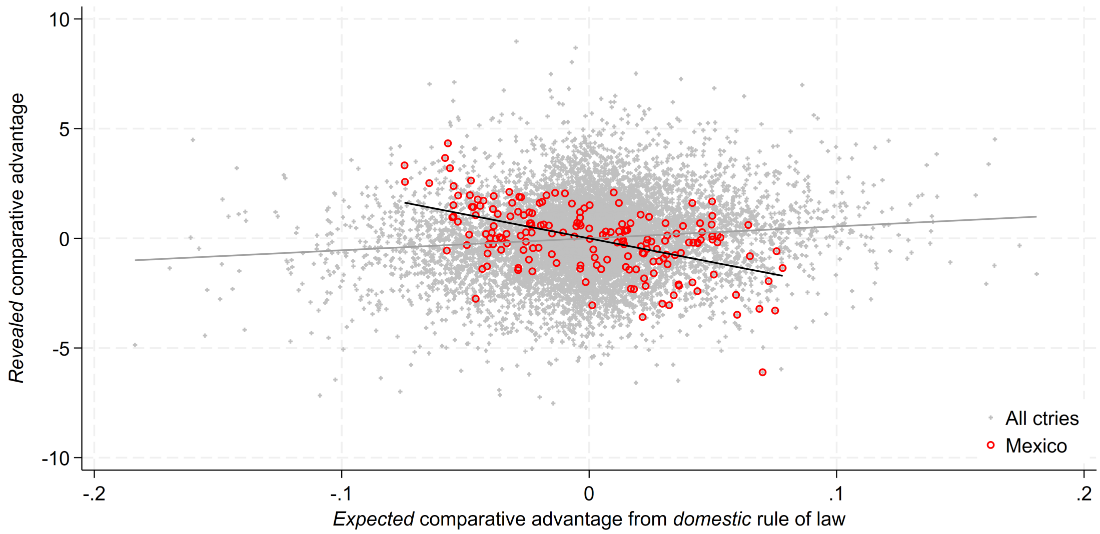
The model developed in this section has three main takeaways. First, it formalizes the idea that contracting distortions raise the cost of doing business, which appears in the model as an increase in the effective price of input quality. Second, it shows that stronger contract-enforcement institutions mitigate these distortions. Third, by allowing suppliers and buyers to operate under different institutional environments, the model identifies an additional benefit of openness: firms can access potentially stronger foreign institutions by importing intermediate goods and services.

3.1. Building blocks

Revenue function. Consider a firm with revenue:

$$R = A_1 q \tag{3}$$

FIGURE 2. Domestic versus foreign contract enforcement and comparative advantage



A. Domestic institutions

B. Foreign institutions

Note: See the note to Figure 1 for details on the construction of the variables, controls, and interpretation of the axes. Panel (a) plots the partial relationship between export specialization and domestic institutional comparative advantage, measured by $z_i \times Q_c$. Panel (b) plots the partial relationship between export specialization and foreign-institutions-based comparative advantage, measured by $z_i \times FQ_c$. Both panels are based on the augmented specification that includes domestic and foreign institutional interactions, factor-proportion interactions, and industry and country fixed effects.

where $A_1 > 0$ is an exogenous demand shifter, such as market size. Revenue is linear in output, consistent with a perfectly elastic demand curve, as under perfect competition.

Technology. Firms combine factors of production and intermediate inputs to produce final goods. There is a continuum of intermediate inputs indexed by $\nu \in [0,1]$. We distinguish between two types of inputs. The first are *generic inputs*: relatively homogeneous goods traded in thick markets that require little or no customization to the buyer's specifications and are not subject to contracting frictions. The second are *specific inputs*: differentiated goods that must be customized to the buyer's specifications. Because the model focuses on specific inputs, we represent them by the composite M. Generic inputs, other factors of production, and exogenous productivity shifters are summarized by the parameter $A_2 > 0$.

Production is described by a four-level technology. The first level relates final output, q, to the composite of specific inputs, M, according to

$$q = A_2 \left(\frac{M}{\alpha}\right)^{\alpha}, \qquad \alpha \in (0,1] \tag{4}$$

where α measures the importance of specific inputs in production. The second level aggregates specific inputs into the composite M:

$$M = \left(\int_0^1 m(\nu)^{\beta} d\nu\right)^{\frac{1}{\beta}}, \qquad \beta \in (0,1) \tag{5}$$

where $1/(1-\beta)$ is the elasticity of substitution across specific inputs. We normalize the quantity of each input to one, so $m(\nu)$ should be interpreted as the *quality* of input ν.

The third level determines input quality. The quality of input ν depends on services provided by the supplier, $a(\nu)$, and services provided by the buyer, $h(\nu)$. We refer to the former as *assembly services* and to the latter as *know-how transfer*. To fix ideas, consider a product manufactured by a Mexican maquiladora for a U.S. firm. The quality of the product depends not only on the skill and effort of the Mexican contractor but also on the know-how transferred by the U.S. buyer. We capture this complementarity with

$$m(\nu) = \left[\frac{a(\nu)}{1-x}\right]^{1-x} \left[\frac{h(\nu)}{x}\right]^{x}, \qquad x \in (0,1) \tag{6}$$

where x measures the importance of know-how transfer for the quality of specific inputs.

The fourth level describes how assembly services and know-how transfer are produced from a continuum of infinitesimal *tasks*, indexed by $\iota \in [0,1]$. Following Acemoglu, Antràs, and Helpman (2007) and Antràs and Helpman (2008), these service aggregates are Cobb-Douglas across tasks:

$$a(\nu) = \exp\left\{\int_0^1 \log a(\nu,\iota) d\iota\right\}$$

$$h(\nu) = \exp\left\{\int_0^1 \log h(\nu, \iota) d\iota\right\}$$

Contractibility. The services provided by both parties are only *partially contractible*: only a subset of the tasks required to provide these services is *verifiable,* in the sense of Hart and Holmström (1987).[5] Let $\mu \in [0, 1]$ denote the share of tasks involved in assembly services that is verifiable, and let $\lambda \in [0, 1]$ denote the corresponding share for know-how transfer. The assembly and know-how aggregates can then be written as

$$a(\nu) = \exp\left\{\underbrace{\int_0^{\mu} \log a(\nu, \iota) d\iota}_{\text{verifiable}} + \underbrace{\int_{\mu}^1 \log a(\nu, \iota) d\iota}_{\text{not verifiable}}\right\}$$

$$h(\nu) = \exp\left\{\underbrace{\int_0^{\lambda} \log h(\nu, \iota) d\iota}_{\text{verifiable}} + \underbrace{\int_{\lambda}^1 \log h(\nu, \iota) d\iota}_{\text{not verifiable}}\right\}$$

Because tasks are symmetric within each service type and verifiability class, there is no reason for a party to choose different task levels within the same group. Let $a_c(\nu)$ and $a_n(\nu)$ denote the levels chosen for verifiable and nonverifiable assembly tasks, respectively. Similarly, let $h_c(\nu)$ and $h_n(\nu)$ denote the levels chosen for verifiable and nonverifiable know-how-transfer tasks. Assembly services and know-how transfer can therefore be written as

$$a(\nu) = a_c(\nu)^{\mu} a_n(\nu)^{1-\mu} \tag{7}$$

$$h(\nu) = h_c(\nu)^{\lambda} h_n(\nu)^{1-\lambda} \tag{8}$$

Costs and relationship-specificity. Performing each task, whether verifiable or nonverifiable, costs $c_a(\nu)$ to the supplier of input ν and c_h to the buyer. The costs of providing assembly services and know-how transfer are therefore

$$C_a(\nu) = c_a(\nu)\left[\mu a_c(\nu) + (1-\mu) a_n(\nu)\right] \tag{9}$$

$$C_h(\nu) = c_h\left[\lambda h_c(\nu) + (1-\lambda) h_n(\nu)\right] \tag{10}$$

[5] Following Grossman and Hart (1986) and Hart and Moore (1990), we assume that the relevant transaction cost is the cost of *describing* the characteristics of what is traded—for example, quality—and the actions the parties must take—for example, their investments—to outside parties. These characteristics and actions may be observable to the contracting parties but not verifiable by outsiders. This creates an informational asymmetry between the parties and the courts known as *lack of verifiability*; see **?**. In principle, verifiability may depend on both input characteristics and the institutional environment. For example, more complex inputs, whose quality is multidimensional and difficult to describe unambiguously, may be less verifiable. I abstract from these input-level differences and focus instead on how stronger contract-enforcement institutions increase verifiability.

Specific inputs are *fully relationship-specific*: their value to alternative buyers is zero. Thus, once these costs are incurred, they are sunk.

Market power. There is an arbitrarily large number of potential suppliers of assembly services. Buyers therefore have full ex-ante market power in this market, which allows them to offer take-it-or-leave-it contracts for the assembly of each input. In addition, all suppliers can provide assembly services for either generic or specific inputs. Since generic inputs are traded in perfectly competitive markets, suppliers of generic inputs earn zero economic profits. It follows that the outside value of suppliers of specific inputs is also zero.

Contracts and payoffs. Contracts can specify four elements: (i) the quantity to be delivered, normalized to one; (ii) an upfront payment $f(\nu)$, paid when the contract is signed; (iii) a delivery payment $s(\nu)$, paid when the input is delivered; and (iv) the levels of verifiable tasks, $a_c(\nu)$ and $h_c(\nu)$, to be completed by the supplier and the buyer, respectively. Given suppliers' outside value, the payoffs of the supplier and the buyer are

$$\pi_a(\nu) = \max\left\{f(\nu) + s(\nu) - C_a(\nu), 0\right\} \tag{11}$$

$$\pi_b = R - \int_0^1 \left[f(\nu) + s(\nu) + C_h(\nu)\right] d\nu \tag{12}$$

3.2. Equilibrium under complete contracts

This section characterizes sourcing when all tasks are verifiable, so that contracts are complete: $\lambda = \mu = 1$. This benchmark will allow us to identify the inefficiencies generated by incomplete contracting in section 3.3. When all tasks are verifiable, contracts can specify and enforce the levels of all tasks. The distinction between verifiable and nonverifiable tasks is therefore irrelevant.[6] Instead of specifying $a_c(\nu)$ and $h_c(\nu)$, contracts specify $a(\nu)$ and $h(\nu)$ directly.

Because generic inputs and other factors of production are absorbed into the productivity term A_2, I characterize the complete-contracting benchmark as a sourcing-cost minimization problem. The buyer chooses the least-cost bundle of supplier and buyer services required to attain a target revenue level $\overline{R}$. Equivalently, since revenue is monotone in output, the buyer minimizes sourcing costs conditional on producing the output level

[6] For example, the parties have incentives to honor the contract if it includes a clause requiring any deviating party to pay an arbitrarily large penalty to the other party. This threat is credible because courts can verify whether a deviation occurred and can therefore enforce the penalty.

associated with $\overline{R}$. The buyer's problem is:

$$\begin{aligned} &\min_{\{a(\nu),h(\nu),s(\nu),f(\nu)\}} \int_0^1 \left[s(\nu) + f(\nu) + C_h(\nu)\right] d\nu \\ \text{s.t.} \quad & s(\nu) + f(\nu) - C_a(\nu) \geq 0, \quad \forall \nu \in [0,1] \\ & R\Big(\big\{a(\nu), h(\nu)\big\}\Big) = \overline{R} \end{aligned}$$

The timing of payments is irrelevant under complete contracts: only the total payment $f(\nu) + s(\nu)$ matters for the supplier. Because the buyer has all *ex-ante* bargaining power, it pays each supplier the minimum amount consistent with participation. Hence, suppliers' participation constraints bind in equilibrium:

$$s(\nu) + f(\nu) = C_a(\nu) = c_a(\nu)a(\nu), \quad \forall \nu \in [0,1]$$

Substituting this condition into the buyer's objective function and using equations 9 and 10, the complete-contracting problem becomes:

$$\begin{aligned} &\min_{\{a(\nu),h(\nu)\}} \int_0^1 \left[c_a(\nu)a(\nu) + c_h h(\nu)\right] d\nu \\ \text{s.t.} \quad & R\Big(\big\{a(\nu), h(\nu)\big\}\Big) = \overline{R} \end{aligned}$$

Since all tasks are contractible, the allocation is efficient: the buyer internalizes the full cost of supplier-provided and buyer-provided services.

Services and input quality. Using equations 3–6, the solution to the complete-contracting cost-minimization problem yields the following levels of assembly services, know-how transfer, and input quality:

$$a^{CC}(\nu) = \left(\frac{1-x}{c_a(\nu)}\right)\left[p(\nu)m^{CC}(\nu)\right] \tag{13}$$

$$h^{CC}(\nu) = \left(\frac{x}{c_h}\right)\left[p(\nu)m^{CC}(\nu)\right] \tag{14}$$

$$m^{CC}(\nu) = \alpha\left[\frac{P}{p(\nu)}\right]^{\frac{1}{1-\beta}}\left(\frac{\overline{R}}{A_1 A_2}\right)^{\frac{1}{\alpha}} \tag{15}$$

Here $p(\nu)$ is the unit cost of quality for input ν, and P is the corresponding ideal price index for the composite of specific inputs, M:

$$p(\nu) \equiv c_a(\nu)^{1-x} c_h^x, \qquad P \equiv \left[\int_0^1 p(\nu)^{-\frac{\beta}{1-\beta}} d\nu\right]^{-\frac{1-\beta}{\beta}}$$

Trade flows. From an empirical perspective, tasks, services, and input quality are not directly observed. It is therefore useful to focus on the recorded transaction value, denoted by $X(\nu)$. Since the costs of know-how transfer are borne internally by the buyer, observed trade flows include only payments to the supplier, $f(\nu) + s(\nu)$. Under complete contracts, these payments equal the supplier's cost of providing assembly services:

$$X^{CC}(\nu) \equiv f(\nu) + s(\nu) = c_a(\nu)a^{CC}(\nu) = (1 - x)\left[p(\nu)m^{CC}(\nu)\right] \tag{16}$$

Thus, $X^{CC}(\nu)$ is a constant share, $1 - x$, of the total cost of producing one unit of input ν with quality $m^{CC}(\nu)$. although observed trade flows include only payments to suppliers and exclude the buyer's internal costs of know-how transfer, they are proportional to the total cost of input quality. This proportionality follows from the Cobb-Douglas structure of input quality in equation 6.

3.3. Equilibrium under incomplete contracts

We now consider the case in which only a fraction of tasks is verifiable. Specifically, a share μ of assembly tasks and a share λ of know-how-transfer tasks are verifiable. As before, contracts specify the quantity to be delivered, normalized to one, and the upfront payment made at the time of signing, $f(\nu)$. Unlike under complete contracts, contracts can now credibly specify only the levels of verifiable tasks, $a_c(\nu)$ and $h_c(\nu)$. They do not specify the delivery payment, $s(\nu)$, in advance. Instead, this payment is *negotiated* after services have been provided but before the input is delivered.

The reason is that contracts cannot make payments contingent on nonverifiable tasks, $a_n(\nu)$ and $h_n(\nu)$. If an unconditional delivery payment $s(\nu)$ were specified in the initial contract, suppliers would have no incentive to choose $a_n(\nu) > 0$: nonverifiable tasks are costly, while the promised payment would be enforceable regardless of whether those tasks were performed. In equilibrium, suppliers would provide no nonverifiable assembly services, and trade in specific inputs would collapse.[7]

Renegotiation. Trade collapse can be avoided if the parties negotiate the delivery payment $s(\nu)$ after investments are made. Although nonverifiable tasks cannot be described to

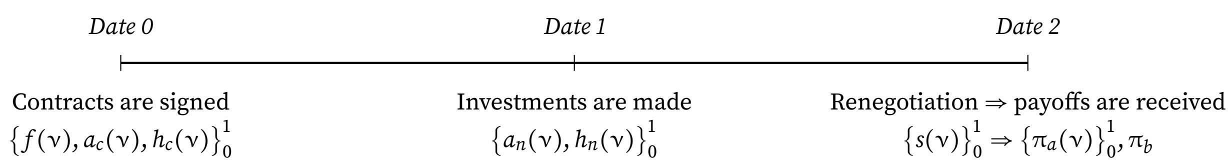
[7]This argument may fail if contracts can condition payments on the *claims* that parties send to the enforcement agency regarding the values of $a_n(\nu)$ and $h_n(\nu)$, even though the actual values are not themselves verifiable. A large literature studies whether lack of verifiability is sufficient to generate efficiency losses when more general contracts are allowed; see Rogerson (1992), Che and Hausch (1999), Tirole (1999), Segal (1999), and Maskin and Tirole (1999). Despite this caveat, Proposition 3 in Che and Hausch (1999) implies that, in an environment with risk-neutral agents, purely cooperative investments, and renegotiation, the first-best allocation cannot be achieved by any contract. I nevertheless cannot rule out the possibility that some feasible contracts, while inefficient, may dominate the renegotiation protocol considered here. To abstract from this possibility, I follow the justification in Acemoglu, Antràs, and Helpman (2007): sophisticated message-contingent mechanisms are incompatible with weak contracting institutions.

outside parties, they are observable to the contracting parties. Therefore, once $a_n(\nu)$ and $h_n(\nu)$ have been chosen, the parties share the same information about the quality of the input. Moreover, because payments and quantities are verifiable, any agreement reached at this stage can be formalized in a second contract specifying that the supplier delivers one unit of the input in exchange for payment $s(\nu)$. This second contract does not need to describe quality, since quality has already been determined by the parties' prior choices.

Rather than explicitly modeling the bargaining process, I follow Acemoglu, Antràs, and Helpman (2007) and impose a bargaining outcome. Specifically, I assume that each supplier is paid its *Shapley value*, a solution concept for cooperative multilateral games.[8]

Timing. There are three periods: dates 0, 1, and 2. Figure 3 summarizes the sequence of events. At date 0, the buyer offers a contract for each input it wants to purchase. Contracts are signed, and upfront transfers $f(\nu)$ are paid. At date 1, all parties choose the verifiable task levels, $a_c(\nu)$ and $h_c(\nu)$, specified in the contract. They also choose the nonverifiable task levels, $a_n(\nu)$ and $h_n(\nu)$, simultaneously and noncooperatively. At date 2, the parties negotiate delivery payments, these payments are immediately enforced, and all payoffs are realized. All monetary values are expressed in date-2 dollars.

FIGURE 3. Sequence of events.



We solve the game by backward induction. First, we characterize date-2 payoffs using the Shapley value. Second, we derive the date-1 choices of nonverifiable tasks, taking anticipated payoffs as given. Third, we characterize the verifiable task levels and upfront payments chosen by the buyer at date 0. Finally, we compare the incomplete-contracting equilibrium with the complete-contracting benchmark.

3.3.1. Rent sharing

The following proposition characterizes the sharing rule used to allocate revenue among the buyer and suppliers. The allocation is based on Shapley values. Because the Shapley

[8]The *Shapley value* is the unique allocation rule satisfying four axioms that jointly characterize an efficient and "fair" distribution of surplus; see Shapley (1953) and Winter (2002). With a finite number of suppliers, a constant share of revenue, as under Nash bargaining, would be sufficient to induce suppliers to choose $a_n(\nu) > 0$, since each supplier would internalize its effect on revenue. With a continuum of infinitesimal suppliers, however, each individual supplier takes aggregate revenue as given, so constant revenue shares do not provide investment incentives. A useful property of the Shapley value is that a supplier's payoff remains increasing in its own investment even in this continuum environment.

value is originally defined for games with a finite number of players, while the present model features a continuum of suppliers, I use the *asymptotic* Shapley value, following Acemoglu, Antràs, and Helpman (2007).

PROPOSITION 1 (Rent sharing).

a. *The supplier of input* ν *receives the following delivery payment:*

$$s(\nu) = \left(\frac{\alpha}{\alpha+\beta}\right)\left[\frac{m(\nu)}{M}\right]^{\beta} R \tag{17}$$

b. *The buyer retains the following revenue:*

$$s_b = \left(\frac{\beta}{\alpha+\beta}\right) R \tag{18}$$

PROOF. See Appendix A. □

The sharing rule has a natural interpretation. The buyer retains a larger share of revenue when specific inputs are easier to substitute, so that β is higher, or when specific inputs are less important in production, so that α is lower.

3.3.2. Tasks, services and input quality

On date 1, the buyer and suppliers choose the levels of nonverifiable tasks, $h_n(\nu)$ and $a_n(\nu)$, respectively, anticipating the date-2 payments characterized in Proposition 1. At date 0, the buyer chooses the contract terms, taking into account how these terms affect the choices made at dates 1 and 2. The buyer minimizes the cost of obtaining the specific inputs needed to attain the target revenue level $\overline{R}$. The buyer's problem can be written as follows:[9]

$$\begin{aligned}
\min_{\{a_c(\nu), h_c(\nu), f(\nu)\}} \quad & \int_0^1 \left[s(\nu) + f(\nu) + C_h(\nu)\right] d\nu \\
\text{s.t.} \quad & s(\nu) + f(\nu) - C_a(\nu) \geq 0, \quad \forall \nu \in [0,1], \\
& s_b\Big(\{a_c(\nu), h_c(\nu)\}\Big) = \left(\frac{\beta}{\alpha+\beta}\right)\overline{R}, \\
& a_n(\nu) \in \arg\max_{\tilde{a}_n(\nu)} \{s(\nu) + f(\nu) - C_a(\nu)\}, \quad \forall \nu \in [0,1], \\
& h_n(\nu) \in \arg\max_{\tilde{h}n(\nu)} \Big\{s_b\Big(\{\tilde{h}_n(\nu)\}\Big) - C_h(\nu)\Big\}, \quad \forall \nu \in [0,1]
\end{aligned}$$

[9] In the supplier's date-1 problem, aggregate revenue is taken as given. This is consistent with each supplier being infinitesimal.

The first constraint is the supplier's participation constraint, as in the complete-contracting benchmark. The second constraint imposes the target revenue requirement in terms of the buyer's retained revenue under the Shapley allocation. The third and fourth constraints are incentive-compatibility constraints: they account for the fact that, at date 1, suppliers and the buyer choose nonverifiable task levels noncooperatively. As under complete contracts, the buyer has all *ex-ante* bargaining power, so each supplier's participation constraint binds in equilibrium:

$$f(\nu) = C_a(\nu) - s(\nu), \quad \forall \nu \in [0,1]$$

Substituting this condition into the buyer's objective function gives

$$\begin{aligned}
\min_{\{a_c(\nu), h_c(\nu)\}} \quad & \int_0^1 \left[C_a(\nu) + C_h(\nu)\right] d\nu \\
\text{s.t.} \quad & s_b\Big(\{a_c(\nu), h_c(\nu)\}\Big) = \left(\frac{\beta}{\alpha+\beta}\right)\overline{R}, \\
& a_n(\nu) \in \arg\max_{\tilde{a}_n(\nu)} \{s(\nu) + f(\nu) - C_a(\nu)\}, \quad \forall \nu \in [0,1], \\
& h_n(\nu) \in \arg\max_{\tilde{h}_n(\nu)} \Big\{s_b\Big(\{\tilde{h}_n(\nu)\}\Big) - C_h(\nu)\Big\}, \quad \forall \nu \in [0,1]
\end{aligned}$$

Although the buyer retains only part of revenue at date 2, it uses its *ex-ante* market power at date 0 to extract the expected surplus through the upfront transfers $f(\nu)$. As a result, the buyer fully appropriates the net social value generated by the transaction. However, because nonverifiable tasks cannot be specified in the initial contract, the buyer cannot directly choose $a_n(\nu)$, and its own choice of $h_n(\nu)$ is distorted by the ex-post sharing rule. The resulting allocation is therefore generally inefficient. The next proposition shows that the equilibrium levels of tasks, services, and input quality under incomplete contracts are proportional to their complete-contracting counterparts.

Proposition 2 (Tasks, services, and input quality).

Let $\psi \equiv x\lambda + (1-x)\mu$ *denote the effective degree of contractibility of input quality, and let*

$$\sigma_b \equiv \frac{s_b}{R} = \frac{\beta}{\alpha+\beta}$$

denote the buyer's share of revenue under the Shapley allocation. Then the incomplete-contracting equilibrium is characterized as follows.

a. *Nonverifiable tasks:*

$$\frac{a_n(\nu)}{a^{CC}(\nu)} = \frac{h_n(\nu)}{h^{CC}(\nu)} = \left\{\sigma_b^{1-\beta\psi}\left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta}\right]^{\beta\psi}\right\}^{\frac{1}{1-\beta}} \equiv (\Gamma_n)^{\frac{1}{1-\beta}} \tag{19}$$

b. *Verifiable tasks:*

$$\text{(20)} \qquad \frac{a_c(\nu)}{a^{CC}(\nu)} = \frac{h_c(\nu)}{h^{CC}(\nu)} = \left\{ \sigma_b^{\beta(1-\psi)} \left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta} \right]^{1-\beta(1-\psi)} \right\}^{\frac{1}{1-\beta}} \equiv (\Gamma_c)^{\frac{1}{1-\beta}}$$

c. *Assembly services and know-how transfer:*

$$\text{(21)} \qquad \frac{a(\nu)}{a^{CC}(\nu)} = \left\{ \sigma_b^{1-(1-\beta)\mu-\beta\psi} \left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta} \right]^{(1-\beta)\mu+\beta\psi} \right\}^{\frac{1}{1-\beta}} = \left(\Gamma_n^{1-\mu} \Gamma_c^{\mu} \right)^{\frac{1}{1-\beta}}$$

$$\text{(22)} \qquad \frac{h(\nu)}{h^{CC}(\nu)} = \left\{ \sigma_b^{1-(1-\beta)\lambda-\beta\psi} \left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta} \right]^{(1-\beta)\lambda+\beta\psi} \right\}^{\frac{1}{1-\beta}} = \left(\Gamma_n^{1-\lambda} \Gamma_c^{\lambda} \right)^{\frac{1}{1-\beta}}$$

d. *Input quality:*

$$\text{(23)} \qquad \frac{m(\nu)}{m^{CC}(\nu)} = \left\{ \sigma_b^{1-\psi} \left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta} \right]^{\psi} \right\}^{\frac{1}{1-\beta}} = \left(\Gamma_n^{1-\psi} \Gamma_c^{\psi} \right)^{\frac{1}{1-\beta}} \equiv (\Gamma)^{\frac{1}{1-\beta}}$$

PROOF. See Appendix A. □

Proposition 2 shows that incomplete contracts affect tasks, services, and input quality through proportional distortion terms with a common structure. Each term is a weighted geometric average of the same two components, differing only in the weights assigned to each component. The first component is the buyer's share of revenue after bargaining, σ_b. The second is:

$$B \equiv \frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta}$$

To interpret these expressions, note that the second component is decreasing in σ_b. Hence, a higher buyer revenue share increases the distortion term through the first component but decreases it through the second. This tension reflects the trade-off between providing incentives to the buyer and providing incentives to suppliers. A higher σ_b allows the buyer to retain a larger share of revenue, which raises input quality by strengthening its incentives to invest in nonverifiable know-how-transfer tasks and to choose higher levels of verifiable tasks when designing the contract. However, a higher σ_b also reduces suppliers' expected payments, weakening their incentives to invest in nonverifiable assembly tasks. Which incentive effect is more important depends on the relative importance of know-how transfer and assembly services for input quality, captured by x, and on the relative contractibility of those services, captured by λ and μ. These forces are summarized by ψ. The next section derives the main implications of incomplete contracts for input provision.

3.3.3. Discussion

Consistent with intuition, incomplete contracts reduce input quality, since $\Gamma \leq 1$. The reason is that parties cannot fully appropriate the returns to their investments in nonverifiable tasks. The distortion operates most directly through nonverifiable tasks, which are undersupplied: $\Gamma_n \leq 1$. However, incomplete contracts also reduce the provision of verifiable tasks. Even though these tasks can be specified and enforced, the buyer chooses them subject to the incentive constraints governing nonverifiable tasks. Thus, verifiable tasks are also undersupplied, $\Gamma_c \leq 1$, although by less than nonverifiable tasks: $\Gamma_n \leq \Gamma_c$. Since both types of tasks are undersupplied, assembly services and know-how transfer are below their complete-contracting levels, resulting in lower-quality specific inputs.

The magnitude of these distortions depends on the model's fundamental parameters. When specific inputs are more important in production, so that α is higher, input quality is more distorted.[10] By contrast, input quality is less distorted when services are more contractible, so that λ or μ is higher, or when inputs are easier to substitute, so that β is higher. If contractibility reflects the quality of contract-enforcement institutions, the model therefore implies that stronger institutions reduce contracting frictions.[11] Proposition 3 formalizes these comparative statics.

PROPOSITION 3 (Effects of incomplete contracts on input quality).

a. *Incomplete contracts lead to underinvestment in all tasks. This distortion is larger for nonverifiable tasks than for verifiable tasks, resulting in lower input quality:*

$$0 \leq \Gamma_n \leq \Gamma_c \leq 1 \quad \Rightarrow \quad \Gamma \in [0, 1]$$

b. *The input-quality distortion term* Γ *is increasing in contractibility, whether of buyer-provided services* (λ) *or supplier-provided services* (μ), *and in the substitutability of specific inputs* (β). *It is decreasing in the importance of specific inputs in production* (α)*:*

$$\Gamma(\overbrace{\alpha}^{-}, \overbrace{\beta}^{+}, \overbrace{\mu}^{+}, \overbrace{\lambda}^{+})$$

c. *Contracting distortions disappear if effective contractibility is complete,* $\psi \to 1$. *This happens if all tasks are fully verifiable,* $\mu \to 1$ *and* $\lambda \to 1$, *or if input quality depends only on the services provided by one party and those services are fully contractible:* $\lambda \to 1$ *and* $x \to 1$, *or*

[10] In this case, the direct effect is to reduce σ_b, which implies that the weakening of buyer incentives dominates.

[11] The role of β combines two effects. On the one hand, a higher β increases σ_b, strengthening incentives for know-how transfer but weakening incentives for assembly services. On the other hand, a higher β also means that inputs are easier to substitute, which reduces suppliers' bargaining power. Since the loss of supplier incentives becomes less consequential when inputs are more substitutable, the net effect is positive.

$\mu \to 1$ *and* $x \to 0$. *Contracting distortions also disappear if specific inputs are unimportant in production,* $\alpha \to 0$. *By contrast, input provision collapses as* $\beta \to 0$*:*

$$\lim_{\psi \nearrow 1} \Gamma = \lim_{\alpha \searrow 0} \Gamma = 1, \qquad \textit{and} \qquad \lim_{\beta \searrow 0} \Gamma = 0$$

PROOF. See Appendix A. □

3.4. From input quality to trade flows

Proposition 2 shows how incomplete contracts distort the sourcing of specific inputs. The game between the buyer and suppliers leads to distortions in the provision of verifiable tasks, nonverifiable tasks, services, and input quality. These distortions are summarized by the terms Γ_c, Γ_n, and Γ. Proposition 3 then shows that these distortions imply under-provision of tasks and lower input quality. While these results are useful, they concern objects—tasks, services, and quality—that are not directly observed in the data. For the model to be useful empirically, we need to derive predictions for observable objects, such as trade flows: the transaction values paid for goods and services.

The model features two transfers to suppliers. The first is the upfront payment, $f(\nu)$, which can be negative. In that case, the supplier pays a fee to win the contract. The second is the delivery payment, $s(\nu)$, which is negotiated after investments are made and paid when the input is delivered. I assume that the observed trade flow, $X(\nu)$, corresponds to the net payment from the buyer to the supplier. Since the supplier's participation constraint binds in equilibrium, this net payment equals the supplier's cost of providing assembly services:

$$X(\nu) = s(\nu) + f(\nu) = c_a(\nu)\Big[\mu a_c(\nu) + (1 - \mu)a_n(\nu)\Big]$$

Under complete contracts, the net payment for input ν is proportional to the total cost of input quality (see 16). When contracts are incomplete, trade flows are equal to their complete-contracting value multiplied by a distortion term that depends on Γ_c and Γ_n:

$$X^{IC}(\nu) = \underbrace{c_a(\nu)a^{CC}(\nu)}_{\substack{\text{trade flow under} \\ \text{complete contracts}}} \underbrace{\left[\mu(\Gamma_c)^{\frac{1}{1-\beta}} + (1 - \mu)(\Gamma_n)^{\frac{1}{1-\beta}}\right]}_{\text{contracting distortion}} \tag{24}$$

Since $0 \leq \Gamma_n \leq \Gamma_c \leq 1$, the distortion term in equation 24 is also between zero and one. Therefore, incomplete contracts reduce observed trade flows relative to the complete-contracting benchmark.

Although contracting frictions do not directly affect the cost of tasks, $c_a(\nu)$ and c_h, and therefore do not change the unit cost of quality, $p(\nu) \equiv c_a(\nu)^{1-x} c_h^x$, equation 24 shows that they affect trade flows in the same way as a mark-up over marginal costs would. To see this, use the definitions of Γ_c, Γ_n, and Γ, to write:

$$\Gamma_c = \Gamma^{\beta} B^{1-\beta}, \qquad \text{and} \qquad \Gamma_n = \Gamma^{\beta} \sigma_b^{1-\beta}$$

Substituting these identities into equation 24 gives:

$$X^{IC}(\nu) = (1-x)\Big[p(\nu) m^{CC}(\nu)\Big] \Gamma^{\frac{\beta}{1-\beta}} \Big[\mu B + (1-\mu)\sigma_b\Big]$$

This expression can be rewritten as if contracting frictions introduced a wedge, Λ, into the unit cost of quality:

$$X^{IC}(\nu) = (1-x) \times \underbrace{\Big[p(\nu)\Lambda\Big]}_{\substack{\text{Effective unit} \\ \text{cost of quality}}} \times \underbrace{\left\{ \alpha \left[\frac{P}{p(\nu)\Lambda}\right]^{\frac{1}{1-\beta}} \left(\frac{\overline{R}}{A_1 A_2}\right)^{\frac{1}{\alpha}} \right\}}_{\substack{\text{Quality demand evaluated at the} \\ \text{the effective cost}}} \tag{25}$$

The wedge Λ is defined by:

$$\Lambda \equiv \Gamma^{-1} \Big[\mu B + (1-\mu)\sigma_b\Big]^{-\frac{1-\beta}{\beta}} \tag{26}$$

Thus, incomplete contracts are isomorphic to a cost wedge on specific input quality. Although contracting frictions do not alter the technological unit cost $p(\nu)$, they reduce the effective provision of quality in the same way that a higher unit cost would. Interestingly, Λ can be characterized in a way that mirrors the characterization of Γ, as shown in the next proposition.

PROPOSITION 4 (Effects of incomplete contracts on unit costs).

a. *Incomplete contracts raise the effective unit cost of input quality:*

$$\Lambda \in [1, +\infty)$$

b. *The effective unit cost of input quality decreases with contractibility, whether of buyer-provided tasks* (λ) *or supplier-provided tasks* (μ). *It increases with the importance of specific inputs in production* (α)*:*

$$\Lambda(\overbrace{\alpha}^{+}, \overbrace{\mu}^{-}, \overbrace{\lambda}^{-})$$

If $\mu \leq \lambda$, *the effective unit cost of input quality also decreases with the elasticity of substitution* *across specific inputs:*

$$\mu \leq \lambda \Rightarrow \frac{d\Lambda}{d\beta} \leq 0$$

c. *Contracting unit cost distortions disappear if effective contractibility is complete,* $\psi \to 1$. *This happens if all tasks are fully verifiable,* $\mu \to 1$ *and* $\lambda \to 1$, *or if input quality depends only on assembly services and they are fully contractible,* $\mu \to 1$ *and* $x \to 0$. *Contracting unit cost distortions also disappear if specific inputs are unimportant in production,* $\alpha \to 0$. *By contrast, effective unit costs become arbitrarily large if specific inputs are essential,* $\beta \to 0$*:*

$$\lim_{\psi \nearrow 1} \Lambda = \lim_{\alpha \searrow 0} \Lambda = 1, \quad \textit{and} \quad \lim_{\beta \searrow 0} \Lambda = +\infty$$

PROOF. See Appendix A. □

The model generates several predictions. First, domestic contract-enforcement institutions affect costs through more than one channel, but existing empirical tests do not distinguish between them. One channel operates through trade costs: contracting frictions raise the effective cost of sourcing specific inputs. This channel should matter primarily for exports of specific inputs. A second channel operates through production costs: weak contract enforcement raises the cost of producing goods that rely heavily on specific inputs, affecting sectors and products to different degrees depending on their technological characteristics. Second, in an open economy, firms may be able to partially bypass weak domestic institutions by importing relationship-specific inputs from countries with stronger contract-enforcement institutions. The empirical results provide evidence for the two cost channels, but not for this bypass channel.

4. Contracting frictions and international trade

This section embeds the results from the previous section into a general equilibrium model of international trade with multiple sectors and input-output linkages, building on Caliendo and Parro (2015). The model distinguishes between trade in intermediate goods and trade in final goods.

The world economy consists of J countries, indexed by n, i, and j, and K sectors, indexed by k and l. Sectors are either tradable, with finite trade costs, or non-tradable, with infinite trade costs. There is a single factor of production, labor, which is perfectly mobile across industries but immobile across countries. All markets are perfectly competitive.

4.1. Consumers

Each country i is populated by an exogenous measure L_i of identical consumers who inelastically supply one unit of labor, without disutility, and consume goods according to a two-tier utility function. In the upper tier, consumers choose consumption across sectors, represented by U_i^k, to maximize a Cobb-Douglas utility function with utility weights $\delta^k \in [0,1]$:

$$U_i = \prod_{k=1}^{K} \left(U_i^k/\delta^k\right)^{\delta^k}$$

Within each sector, there is a unit measure of goods indexed by ω. Consumers choose consumption of each good, $u_i^k(\omega)$, to maximize a CES aggregator with elasticity of substitution $\frac{1}{1-\rho^k} > 1$:

$$U_i^k = \left(\int_0^1 u_i^k(\omega)^{\rho^k} d\omega\right)^{\frac{1}{\rho^k}}$$

These preferences imply the following expressions for expenditure on final good ω and the sector-k price index:

$$X_i^{k,f}(\omega) = \left[p_i^k(\omega)/P_i^{k,f}\right]^{-\frac{\rho^k}{1-\rho^k}} \delta^k w_i L_i$$

$$P_i^{k,f} = \left(\int_0^1 p_i^k(\omega)^{-\frac{\rho^k}{1-\rho^k}} d\omega\right)^{-\frac{1-\rho^k}{\rho^k}} \tag{27}$$

where $p_i^k(\omega)$ is the effective price paid by consumers in country i for final good ω, and $w_i L_i$ is country i's GDP.

4.2. Firms

Each good is produced by a single firm; hence, I use "good" and "firm" interchangeably. Production is represented by the following Cobb-Douglas technology, which combines labor, $L_n^k(\omega)$, and intermediate inputs. Intermediate inputs are represented by input *composites* $M_n^{lk}(\omega)$, one for each upstream sector l:

$$q_n^k(\omega) = \varphi_n^k(\omega)\left[\frac{L_n^k(\omega)}{1-\gamma^k}\right]^{1-\gamma^k} \prod_{l=1}^{K}\left[\frac{M_n^{lk}(\omega)}{\gamma^{lk}}\right]^{\gamma^{lk}} \tag{28}$$

The exponents $\gamma^{lk} \in [0,1]$ and $1-\gamma^k \in [0,1]$ are the output elasticities of composite l and labor, respectively, with $\gamma^k \equiv \sum_l \gamma^{lk}$. All firms in sector k combine labor and composites according to equation 28, but they differ in productivity, represented by $\varphi_n^k(\omega)$. Under these assumptions, cost minimization implies the following constant marginal cost:

$$c_n^k(\omega) = \frac{(w_n)^{1-\gamma^k}\prod_{l=1}^{K}(\tilde{P}_n^l)^{\gamma^{lk}}}{\varphi_n^k(\omega)} = \frac{c_n^k}{\varphi_n^k(\omega)} \tag{29}$$

where $\tilde{P}_n^l$ is the price of one unit of sector l's composite, and $c_n^k \equiv (w_n)^{1-\gamma^k} \prod_{l=1}^K (\tilde{P}_n^l)^{\gamma^{lk}}$ is the component of marginal cost common to all sector-k goods produced in country n.

Firms can sell their products to any country in the world. In each destination, there are two types of buyers. Goods sold to representative consumers are *final goods*. Goods sold to the agents described next are intermediate goods, or simply *inputs*.

4.3. Intermediaries

Each country n and sector k has one *intermediary* that purchases inputs from firms around the world, combines them to produce the sectoral composite, and sells this composite to *domestic* firms at marginal cost. Intermediaries combine intermediate goods using a two-tier production function. In the first tier, they choose the value of generic-input composites, $M_n^{k,g}$, and specific-input composites, $M_n^{k,s}$, to minimize the cost of producing the sectoral composite M_n^k, according to the following Cobb-Douglas function.

$$M_n^k = \left(\frac{M_n^{k,s}}{\alpha^k}\right)^{\alpha^k} \left(\frac{M_n^{k,g}}{1-\alpha^k}\right)^{1-\alpha^k}$$

where $\alpha^k \in [0,1]$ is the composite elasticity of specific inputs. Within each input group, there is a unit measure of inputs indexed by ν. Intermediaries choose purchases of each input to minimize the cost of producing composite $M_n^{k,z}$, where $z = g$ for generic inputs and $z = s$ for specific inputs. Each composite aggregates individual inputs according to the following CES function with elasticity of substitution $\frac{1}{1-\beta^k} > 1$:

$$M_n^{k,z} = \left(\int_0^1 m_n^{k,z}(\nu)^{\beta^k} d\nu\right)^{\frac{1}{\beta^k}} \quad \text{for } z \in \{s, g\}$$

where, as in section 3, quantities are normalized to one, so $m_n^{k,z}(\nu)$ is interpreted as the quality of input ν. Intermediate goods are purchased from the lowest-cost suppliers worldwide. Cost minimization implies the following price indices for $M_n^{k,g}$, $M_n^{k,s}$, and M_n^k:

$$P_n^{k,z} = \left(\int_0^1 p_n^{k,z}(\nu)^{-\frac{\beta^k}{1-\beta^k}} d\nu\right)^{-\frac{1-\beta^k}{\beta^k}} \quad \text{for } z \in \{s, g\} \tag{30}$$

$$\tilde{P}_n^k = \left(P_n^{k,s}\right)^{\alpha^k} \left(P_n^{k,g}\right)^{1-\alpha^k} \tag{31}$$

4.4. Contracting frictions when sourcing specific inputs

The framework developed in section 3 can be used to model the purchase of specific inputs. In this application, intermediaries are the buyers. Because they sell composites to individual producers at marginal cost, their behavior is equivalent to that of a mass of identical intermediaries facing a horizontal demand curve.[12] Thus, the revenue function of an intermediary can be written as a special case of equations 3, 4 and 5:

$$R_n^k = \underbrace{\tilde{P}_n^k}_{A_1} \overbrace{\underbrace{\left(\frac{M_n^{k,g}}{1-\alpha^k}\right)^{1-\alpha^k}}_{A_2} \left(\frac{M_n^{k,s}}{\alpha^k}\right)^{\alpha^k}}^{q}$$

Therefore, the results from section 3 imply that the price of a specific input sourced from country j can be written as the complete-contracting price amplified by the distortion caused by incomplete contracts: $p_{jn}^{k,s}(\nu) = p_{jn}^k(\nu)\Lambda_{jn}^k$, where Λ_{jn}^k is defined as in equation 26:

$$\Lambda_{jn}^k \equiv (\Gamma_{jn}^k)^{-1} \left\{ \mu_j \left[\frac{1-\sigma_b^k(1-\psi_{jn}^k)\beta^k}{1-(1-\psi_{jn}^k)\beta^k} \right] + (1-\mu_j)\sigma_b^k \right\}^{-\frac{1-\beta^k}{\beta^k}} \tag{32}$$

Here, μ_j is indexed by j because it captures the quality of contract-enforcement institutions in the country from which the input is sourced. Similarly, $\psi_{jn}^k \equiv x^k\lambda_n + (1-x^k)\mu_j$ is indexed by both countries and by sector because it is a linear combination of the exporter institutional-quality parameter, μ_j, and the importer institutional-quality parameter, λ_n, with sector-specific weight x^k. With this definition of prices, expenditure on generic and specific input ν can be written, respectively, as:

$$X_i^{k,g}(\nu) = \left[p_i^k(\nu)/P_i^{k,g}\right]^{-\frac{\beta^k}{1-\beta^k}} (1-\alpha^k)R_i^k$$

$$X_i^{k,s}(\nu) = (1-x^k)\left[p_i^k(\nu)\Lambda_{jn}^k/P_i^{k,s}\right]^{-\frac{\beta^k}{1-\beta^k}} \alpha^k R_i^k$$

where R_i^k is total expenditure by all sectors in country i on sector-k intermediate goods. Note that any observed trade flow of specific inputs necessarily implies $x^k < 1$.

[12] Intermediaries nevertheless have full market power in input markets, so they write take-it-or-leave-it contracts as in section 3.

4.5. International trade, costs and prices

Trade costs are modeled as exogenous iceberg frictions: for each unit sold in market i, a seller in country n must ship $\tau_{ni} > 1$ units, with $\tau_{ni} - 1$ units lost in transit. These frictions satisfy the triangle inequality, so no indirect route through a third country is cheaper than direct trade. Domestic trade costs are normalized to one, so $\tau_{ii} = 1$.

The unit cost, inclusive of trade costs, of a good produced in country n and sold in market i is $c^k_{ni}(\omega) = \frac{c^k_n \tau_{ni}}{\varphi^k_n(\omega)}$. Thus, goods from the same country and sector differ only in productivity, $\varphi^k_n(\omega)$. Following Eaton and Kortum (2002), productivities are random draws from a Fréchet distribution with shape parameter θ and scale parameter T^k_n:

$$\Pr\left(\varphi^k_n(\omega) \leq \varphi\right) = \exp\left\{-T^k_n \varphi^{-\theta}\right\}$$

Countries may differ in fundamental productivity across sectors, represented by the scale parameter T^k_n. As in Costinot, Donaldson, and Komunjer (2012), this parameter captures factors such as climate, infrastructure, and institutions that affect the productivity of all producers in a given country and industry.[13]

Perfect competition implies that suppliers charge marginal cost inclusive of trade costs. Since consumers and intermediaries source goods from the cheapest supplier, the effective equilibrium price of a final good or generic input ω is the lowest unit cost inclusive of trade costs:

$$p^k_n(\omega) = \min_{1 \leq j \leq J} \left\{c^k_{jn}(\omega)\right\} = \min_{1 \leq j \leq J} \left\{\frac{c^k_j \tau_{jn}}{\varphi^k_j(\omega)}\right\}$$

The effective equilibrium price of a specific input ν is the lowest unit cost inclusive of both trade costs and contracting frictions:

$$p^{k,s}_n(\nu) = \min_{1 \leq j \leq J} \left\{p^k_{jn}(\nu) \Lambda^k_{jn}\right\} = \min_{1 \leq j \leq J} \left\{\frac{c^k_j \tau_{jn} \Lambda^k_{jn}}{\varphi^k_j(\nu)}\right\}$$

The assumption that productivities are random draws from a Fréchet distribution implies that prices are random draws from a Weibull distribution and that price indices can be written in terms of fundamental parameters. The price indices for final goods and generic

[13] T^k_n does <u>not</u> capture the *differential* effect of contract-enforcement institutions on specific inputs relative to generic inputs or final goods. However, it can capture effects that are common across different types of goods, as well as the effects of other institutions, such as those pertaining property rights (see Acemoglu and Johnson 2005).

inputs have the same structure, differing only in the level shifters B^k and $\tilde{B}^k$.

$$P_n^{k,f} = B^k \left[\sum_{j=1}^{J} T_j^k (c_j^k \tau_{jn})^{-\theta} \right]^{-\frac{1}{\theta}} \quad \text{and} \quad P_n^{k,g} = \tilde{B}^k \left[\sum_{j=1}^{J} T_j^k (c_j^k \tau_{jn})^{-\theta} \right]^{-\frac{1}{\theta}} \tag{33}$$

where $B^k \equiv \Gamma\left[\frac{\rho^k+\theta(1-\rho^k)}{\theta(1-\rho^k)}\right]^{-\frac{1-\rho^k}{\rho^k}}$ and $\tilde{B}^k \equiv \Gamma\left[\frac{\beta^k+\theta(1-\beta^k)}{\theta(1-\beta^k)}\right]^{-\frac{1-\beta^k}{\beta^k}}$. The price index for specific inputs has the same structure as the price index for generic inputs, except that it also incorporates the contracting friction from equation 32:

$$P_n^{k,s} = \tilde{B}^k \left[\sum_{j=1}^{J} T_j^k (c_j^k \tau_{jn} \Lambda_{jn}^k)^{-\theta} \right]^{-\frac{1}{\theta}} \tag{34}$$

4.6. Expenditure shares

In addition to price indices, the Eaton and Kortum framework provides expressions for bilateral exports in terms of fundamental parameters. Let $X_{ni}^{k,f} \equiv \int_{\omega \in \Omega_{ni}^{k,f}} X_i^{k,f}(\omega) d\omega$ denote the value of total exports of final goods from country n to country i in sector k, where $\Omega_{ni}^{k,f} \equiv \left\{\omega \in [0,1] \mid p_{ni}^k(\omega) = p_i^k(\omega)\right\}$ is the set of final goods in sector k exported by country n to country i. Then:

$$X_{ni}^{k,f} = \underbrace{\frac{T_n^k (c_n^k \tau_{ni})^{-\theta}}{\sum_{j=1}^{J} T_j^k (c_j^k \tau_{ji})^{-\theta}}}_{\pi_{ni}^{k,f}} \delta^k w_i L_i \tag{35}$$

where $\pi_{ni}^{k,f}$ is the probability that country n is the least-cost provider of a final good. Similarly, let $X_{ni}^{k,g} \equiv \int_{\omega \in \Omega_{ni}^{k,g}} X_i^{k,g}(\nu) d\nu$ denote the value of total exports of generic inputs from country n to country i in sector k, where $\Omega_{ni}^{k,g} \equiv \left\{\nu \in [0,1] \mid p_{ni}^k(\nu) = p_i^k(\nu)\right\}$ is the set of generic inputs in sector k exported by country n to country i. Then:

$$X_{ni}^{k,g} = \underbrace{\frac{T_n^k (c_n^k \tau_{ni})^{-\theta}}{\sum_{j=1}^{J} T_j^k (c_j^k \tau_{ji})^{-\theta}}}_{\pi_{ni}^{k,g}} (1-\alpha^k) R_i^k \tag{36}$$

where $\pi_{ni}^{k,g}$ is the probability that country n is the least-cost provider of a generic input. Given the assumptions above, this probability is the same as $\pi_{ni}^{k,f}$. Finally, let $X_{ni}^{k,s} \equiv \int_{\omega \in \Omega_{ni}^{k,s}} X_i^{k,s}(\nu) d\nu$ denote the value of total exports of specific inputs from country n to country i in sector k, where $\Omega_{ni}^{k,s} \equiv \left\{ \nu \in [0,1] \mid p_{ni}^{k}(\nu)\Lambda_{ni}^{k} = p_{i}^{k,s}(\nu) \right\}$ is the set of specific inputs in sector k exported by country n to country i. Then:

$$X_{ni}^{k,s} = \underbrace{\frac{T_n^k (c_n^k \tau_{ni} \Lambda_{ni}^k)^{-\theta}}{\sum_{j=1}^{J} T_j^k (c_j^k \tau_{ji} \Lambda_{ji}^k)^{-\theta}}}_{\pi_{ni}^{k,s}} \alpha^k (1 - x^k) R_i^k \tag{37}$$

where $\pi_{ni}^{k,s}$ is the probability that country n is the least-cost provider of a specific input.

4.7. Discussion

Equations 35-37 have several implications for the relationship between trade and institutions and for how this relationship should be tested empirically.

Contracting frictions affect trade flows through two channels. The first, which we call the **trade cost channel**, is represented by Λ_{ni}^{k} in equation 37. This channel operates by inflating prices conditional on the production cost c_n^k. Because it is present only when specific inputs are traded, it highlights the importance of distinguishing specific inputs from other types of goods in empirical work.[14]

The second, which we call the **production cost channel**, is represented by the price index of specific inputs, $P_n^{k,s}$, in equation 34. This price index is a component of the input price index $\tilde{P}_n^k$ in equation 31, which in turn enters the factory-gate unit cost $c_n^k \equiv (w_n)^{1-\gamma^k} \prod_{l=1}^{K} (\tilde{P}_n^l)^{\gamma^{lk}}$. Thus, goods produced in sectors that rely more intensively on specific inputs, that is, sectors with high $\gamma^{lk}\alpha^l$, are more sensitive to the quality of contract-enforcement institutions. In contrast to the trade cost channel, this channel affects trade flows for all types of goods. It therefore highlights the importance of controlling for cross-sector differences in the role of specific inputs in production. To my knowledge, the distinction between these two channels has not been explicitly acknowledged in the literature.[15]

[14] Alternatively, one could test the weaker prediction that contract-enforcement institutions have a *stronger effect* on trade in specific inputs than on trade in other goods. This may be more reasonable given that institutions can affect trade through channels unrelated to relationship-specific investments and the hold-up problem. For example, Schmidt-Eisenlohr (2013) develops a model in which contractual risks arise from the time gap between production and sale. In that setting, there is no need to distinguish between final and intermediate goods, much less between specific and generic inputs, unless one is willing to assume, and able to measure, systematic differences across goods in the time gap between production and sale.

[15] For example, Berkowitz, Moenius, and Pistor (2006) compares the effect of institutional quality on bilateral trade in "complex" versus "simple" industries, where *complexity* is measured directly using the classification in Rauch (1999). This approach therefore constitutes a test of the trade cost channel, since it focuses on whether the industry itself is specific. In contrast, Nunn (2007) uses a measure of *contract intensity*, defined as the proportion of intermediate inputs that are relationship-specific, where relationship specificity is also measured using the classification in Rauch (1999). Because the emphasis is not on the specificity of

A country with weak institutions may export specific inputs for which know-how transfers are important. If specific inputs require relationship-specific investments by *both* parties, the institutions of the importing country also matter. Proposition 4 shows that the importance of the importer's institutions is greater for specific inputs for which know-how transfers are more important, whereas the importance of the exporter's institutions is greater for specific inputs for which assembly services are more important. Hence, a firm located in a country with weak institutions may still be able to export specific inputs to countries with strong institutions if those inputs are intensive in know-how transfers. This prediction can be tested empirically by controlling for the relative importance of buyer and supplier services.[16]

A country with weak institutions may export contract-intensive goods if it imports its specific inputs. Equation 34 shows that, in an open economy, the price index for specific inputs depends on the contracting frictions associated with all potential supplier countries. Thus, a firm may partially bypass weak domestic institutions by importing specific inputs from countries with stronger institutions. This highlights the importance of accounting for *input tradability* in empirical work. This mechanism has been largely overlooked in the literature, which tends to focus on domestic institutions, as in Nunn (2007), Levchenko (2007), and Costinot (2009).

A theoretically consistent test of contracting frictions requires bilateral trade data. Because contracting frictions may depend on the contract-enforcement institutions of both importing and exporting countries, and because the trade cost and production cost channels operate differently, the empirical analysis should be based on gravity regressions using bilateral trade data. This approach makes it possible to distinguish the role of importer institutions from that of exporter institutions and to test the two channels separately. It also contrasts with the use of aggregate export data in Nunn (2007) and Levchenko (2007).

5. Data and variable definitions

Trade flows. I use bilateral product-level trade flows for 2012, 2015, and 2018 from the *Base pour l'Analyse du Commerce International* (BACI, see Gaulier and Zignago 2010), a dataset maintained by the *Centre d'études prospectives et d'informations internationales* (CEPII) publicly available through its website.[17] BACI is updated yearly, and different versions are

the industry itself, but on the specificity of its upstream industries, this approach constitutes a test of the production cost channel.

[16] The literature has used different proxies for these investments. For instance, Antràs (2003) uses physical capital intensity, human capital intensity, R&D intensity, and advertising intensity as proxies.

[17] The information in BACI is based on official trade data reported by countries to the United Nations and disseminated through the *Commodities Trade Statistics* (COMTRADE) database. Because countries report both

identified by their release year and month. I use the January 2023 version. In BACI, a product is defined as a 6-digit Harmonized System (HS) subheading, revision 2012.[18]

Sector classification. I assign products to economic sectors using the conversion key from the OECD's *Bilateral Trade Database by Industry and End-Use* (BTDIxE), which maps each product, defined as in BACI, to a sector in the fourth revision of the United Nations' *International Standard Industrial Classification of all Economic Activities* (ISIC). I keep products classified as "Agriculture, Forestry and Fishing", "Mining and Quarrying", or "Manufacturing", which together account for more than 98% of products and trade value in the data. Table 1 lists the 18 sectors included in the sample and reports the distribution of products and trade value across them. The three largest sectors by trade value are "Computer, electronic and optical goods" (14%), "Mining and quarry" (11%), and "Motor vehicles, trailers and semi-trailers" (9%).

Product classification. We classify products in BACI as final goods, specific inputs, or generic inputs using the fifth revision of Broad Economic Categories (BEC) from 2016. In this revision, products are defined at the 6-digit HS level using the 2012 HS revision, making BEC compatible with BACI. The initial sample contains 5,120 product codes. BEC provides a hierarchical classification of products organized into six levels, or "dimensions." I use the third, fourth, and fifth dimensions. The third dimension classifies products by *end use* as intermediate consumption, gross fixed capital formation, or final consumption. The fourth dimension classifies products by *processing stage* as primary or processed. The fifth dimension classifies products as *generic or specific*.

I map final-consumption goods to *final goods* in the model and intermediate-consumption goods to *intermediate inputs*. Within intermediate inputs, goods not classified as specific, namely those classified as generic or primary, are treated as *generic inputs*. Some products have dual classifications. I resolve these cases by reclassifying some products and dropping others, leaving 4,997 products in the final product-level sample. Table 2 summarizes the distribution of products and trade value across the final categories.

Trade is distributed relatively evenly across final goods, generic inputs, and specific inputs. Final goods account for 29.8% of trade value, generic inputs for 28.8%, and specific inputs for 28.6%. The remaining 12.8% corresponds to capital goods, which are reported in the table but excluded from the analysis sample.

imports and exports, bilateral trade flows are often reported twice in the raw data. Although these values should coincide, in practice they often differ. CEPII therefore applies a harmonization procedure to reconcile duplicate and inconsistent reports into a single trade-flow measure. BACI is the result of this procedure.

[18]The HS is a standard classification system for goods used by most customs authorities. Since its creation in 1988, the HS has been revised several times, with each revision identified by the year of its introduction.

TABLE 1. Distribution of products and trade across economic sectors.

Sector		Products		Trade value	
Code	Name	Nbr.	%	bill. US$	%
A	Agriculture, forestry and fishing	368	7	1,592	3
B	Mining and quarrying	93	2	5,611	11
C10_12	Food products, beverages and tobacco	571	11	3,104	6
C13_15	Textiles, textile products, leather and footwear	824	16	2,830	5
C16_18	Wood, paper products and printing	207	4	1,066	2
C19	Coke and refined petroleum products	17	<1	2,604	5
C20	Chemicals and chemical products	810	16	4,392	8
C21	Pharmaceutical products	75	2	1,614	3
C22	Rubber and plastics products	134	3	1,331	3
C23	Other non-metallic mineral products	164	3	580	1
C24	Basic metals	373	7	3,681	7
C25	Fabricated metal products	237	5	1,307	3
C26	Computer, electronic and optical products	269	5	7,518	14
C27	Electrical equipment	177	4	2,525	5
C28	Machinery and equipment n.e.c.	466	9	4,097	8
C29	Motor vehicles, trailers and semi-trailers	64	1	4,541	9
C30	Other transport equipment	80	2	1,794	3
C31_32	Furniture; other manufacturing	191	4	1,954	4

Note: The trade statistics come from the BACI dataset for years 2012, 2015 and 2018. Products are assigned a sector using the conversion key from the OECD's BTDIxE.

TABLE 2. Distribution of products and trade across classifications.

BEC classification			Products		Trade value		Final
End-use	Processing	Specification	Nbr.	%	bill. US$	%	classification
Final consumption			1,356	27	14,776	30	Final goods
Intermediate consumption	Primary		329	7	5,728	12	Generic inputs
	Processed	Generic	1,096	22	8,548	17	
		Specific	1,631	33	14,169	29	Specific inputs
Gross Fixed Capital Formation			585	12	6,365	13	Not included

Note: The trade statistics come from the BACI dataset for years 2012, 2015 and 2018. Products are classified using the fifth revision of BEC.

Country variables. For each country, we construct measures of physical capital endowments, human capital endowments, and contract-enforcement quality. **Physical capital** endowment, K_{nt}, is defined as the log of capital stock per worker, measured in 2017 Purchasing Power Parity (PPP) U.S. dollars, and comes from version 10.01 of the *Penn World Table* (PWT), available from the Groningen Growth and Development Centre website.

The **human capital** index, H_{nt}, comes from the same source and is defined as the log of the relative income of a worker with average years of schooling relative to an otherwise identical worker with no schooling.[19]

Finally, **contract-enforcement quality**, μ_{nt} and λ_{nt}, is measured using the "Rule of Law" index from the *Worldwide Governance Indicators* (WGI) database (see Kaufmann and Kraay 2022).[20] The original index is not bounded between 0 and 1, so we normalized it. Some countries are missing information for one or more of these variables. Depending on the specification, observations involving these countries are dropped from the sample. However, 75% of country pairs, across all products, have information on all exporter variables and on importer contract-enforcement quality; these observations account for 99% of trade value.

Sector-level variables. For each sector, we construct measures of capital intensity, skill intensity, and R&D intensity using mostly OECD data; details are provided in Appendix **??**. We measure **capital intensity**, cap^k_{nt}, as the ratio of the cost of capital services to total cost, where total cost is the sum of capital-service costs, labor-service costs, and intermediate consumption in the corresponding sector. We measure **skill intensity**, skill^k_{nt}, as the ratio of payroll paid to workers with more than a high school education to total cost in the corresponding sector. We measure **R&D intensity**, rd^k_{nt}, as the ratio of R&D expenditure to gross value added. In all cases, the sector-level measure is the average across countries and years. Finally, the share of specific inputs in *aggregate expenditure* on sector-k inputs, $\boldsymbol{\alpha^k}$, is not directly observed. We therefore proxy for it using the share of specific inputs in the *number* of sector-k intermediate goods, based on our product classification. This follows the logic of Nunn (2007), who uses the share of differentiated inputs to construct his measure of contract intensity. Table 3 reports the three most and least intensive sectors for each indicator.

Sectors producing more specific goods, as measured by the share of specific inputs in the product mix, tend to be slightly less capital intensive, with a correlation of −0.12, and more skill intensive, with a correlation of 0.37. They also tend to have high R&D intensity, with a correlation of 0.62. In turn, sectors with high R&D intensity tend to be skill intensive, with a correlation of 0.52, and somewhat capital intensive, with a correlation of 0.27. Capital intensity is negatively associated with the production of specific inputs and only weakly correlated with skill and R&D intensity, with correlations of 0.24 and 0.26, respectively. Finally, skill intensity is positively correlated with all other indicators.

[19]The human capital of a worker with s years of schooling is defined as relative income compared with an otherwise identical worker with no schooling. Mathematically, it can be written as $H(s) \equiv e^{\psi(s)}$, where the income of a worker with zero schooling is normalized to one if $\psi(0) = 0$. The function $\psi(s)$ is assumed to be piecewise linear in s, reflecting higher returns to earlier stages of schooling.

[20]the Rule of Law index is a perception-based indicator intended to reflect the views of survey respondents, non-governmental organizations, commercial businesses, and public-sector organizations.

TABLE 3. Top and bottom 3 sectors for capital, skill and R&D intensities and for share of specific inputs

	Capital intensity		Skill intensity	
	Code	Name	Code	Name
Most	B	Mining and quarrying	C31_32	Furniture; other manufacturing
	A	Agriculture, forestry and fishing	C21	Pharmaceutical products
	C26	Computer, electronic and optical products	C26	Computer, electronic and optical products
Least	C25	Fabricated metal products, except machinery and equipment	C24	Basic metals
	C10_12	Food products, beverages and tobacco	A	Agriculture, forestry and fishing
	C19	Coke and refined petroleum products	C19	Coke and refined petroleum products
	R&D intensity		**Share of specific inputs in product mix**	
	Code	Name	Code	Name
Most	C26	Computer, electronic and optical products	C21	Pharmaceutical products
	C30	Other transport equipment	C29	Motor vehicles, trailers and semi-trailers
	C21	Pharmaceutical products	C30	Other transport equipment
Least	C16_18	Wood, paper products and printing	C10_12*	Food products, beverages and tobacco*
	B	Mining and quarrying	B*	Mining and quarrying*
	A	Agriculture, forestry and fishing	A*	Agriculture, forestry and fishing*

Note: Products are assigned a sector using the conversion key from the OECD's BTDIxE. Data from OECD's *Structural Analysis* (STAN), *Analytical Business Enterprise Research and Development* (ANBERD); and *American Community Survey* (ACS).
(*) Three sectors are tied in having no specific inputs in their product mix: "Agriculture, forestry and fishing", "Mining and quarrying" and "Food products, beverages and tobacco".

Country-industry variables. We construct variables at the country-industry level using data from the OECD's *Inter-Country Input-Output* (ICIO) of countries, the share of specific inputs in the *output* of each sector (α^k), and a measure for the importance of know-how transfers (x^k).

I follow Antràs (2003) in using both human capital intensity and R&D intensity as proxies.[21] It is important to note that the relevant share of know-how transfers is that of the downstream industry k, not the upstream one l. In the model, by assuming that trade is conducted by intermediaries and that they belong to the same sector as the goods they are buying, this distinction between upstream and downstream is erased. However, when testing the model predictions with data, we account for it.

6. Empirical analysis

In this section, we put to the test the predictions described in section 4.7. From now on, *type* is a categorical variable that indicates whether a commodity is a final good, a generic input, or a specific input; and *RS* is a dummy variable that indicates that the commodity is a specific input. To reduce the dimensionality of the dataset, we aggregated products into industry-type bins.

[21] "To the extent that final-good [downstream] producers also contribute to their supplier's cost related to the acquisition of human capital (e.g., by financing training programs), a model along the lines of the one developed above would indeed predict an effect of human-capital intensity (...) A similar argument could be used to defend the inclusion of some measure of the importance of R&D (...) in the production process." (Antràs 2003, p. 1405).

6.1. Trade cost channel

6.1.1. Empirical strategy

The model implies that incomplete contracts affect trade flows by increasing the effective cost of sourcing relationship-specific inputs. Specifically, when a firm in country n ("destination") sources a specific input from country j ("origin") in industry k, contractual frictions enter the delivered cost of the input through the iceberg wedge Λ_{jn}^{k}. This wedge does not apply to final goods or to generic inputs. The empirical strategy therefore exploits variation across product types within origin–destination–industry–year cells: the model predicts that institutions should affect trade in specific inputs, but not trade in final goods or generic inputs.

Because Λ_{jn}^{k} enters the gravity equation as an iceberg trade cost, the relevant object for the empirical specification is $\ell_{jn}^{k} \equiv \log \Lambda_{jn}^{k}$. The theoretical expression for ℓ_{jn}^{k} is nonlinear in origin institutions, destination institutions, and industry characteristics. Rather than estimate the nonlinear expression directly, we use a first-order approximation that preserves the key economic mechanism of the model.

$$\ell_{jn}^{k} \approx \ell_0 + \ell_\mu \mu_j + \ell_\lambda \lambda_n + \ell_{\lambda x} \left(x^k \times \lambda_n \right) + \ell_{\mu\lambda} \left(\mu_j \times \lambda_n \right) \tag{38}$$

This approximation shares several of the predictions found in proposition 4 about the relationship between the contracting frictions term Λ_{jn}^{k} and the quality of the contract enforcement institutions in origin and destination.

First, better contract enforcement institutions in either country mitigate contracting frictions:

$$\ell_\mu \leq 0 \qquad \text{and} \qquad \ell_\lambda \leq 0$$

Second, contracting frictions are more sensitive to the quality of destination institutions when know-how transfers are more important:

$$\ell_{\lambda x} \leq 0$$

Third, μ_j and λ_n are "substitutes" in the sense that when one is higher, the sensitivity of contracting frictions to the other decreases:

$$\ell_{\mu\lambda} \geq 0$$

We use a battery of fixed effects (indicated below by ζ) to control for other determinants of trade flows. Augmenting our notation to account for year (t) and product type (z), taking logs on both sides in equations 35—37, adding an indicator function for specific inputs

(I_s), and unpacking ℓ^k_{jn} using equation 38, the regression equation can be written as:

$$\begin{aligned}\log X^{kz}_{jnt} &= \zeta^k_{jt} + \zeta^{kz}_{nt} + \zeta_{jnt} - \theta\left(\ell^k_{jnt} \times I_s\right) + \varepsilon^{kz}_{jnt} \\ &\approx \zeta^k_{jt} + \zeta^{kz}_{nt} + \zeta_{jnt} + I_s\Big[\beta_\mu \mu_{jt} + \beta_{\mu\lambda}\left(\mu_{jt} \times \lambda_{nt}\right)\Big] + \varepsilon^{kz}_{jnt}\end{aligned} \tag{39}$$

Here, ζ^k_{jt} are origin–industry–year fixed effects, ζ^{kz}_{nt} are destination–industry–year–type fixed effects, ζ_{jnt} are origin–destination–year fixed effects. These fixed effects absorb a rich set of alternative determinants of trade flows: origin comparative advantage within industries over time, destination demand or input demand within industries over time, common industry-specific shocks by product type, and bilateral trade costs that vary over time. The first-order approximation implies:

$$\beta_\mu \equiv -\theta\ell_\mu \geq 0, \qquad \beta_{\mu\lambda} = -\theta\ell_{\mu\lambda} \leq 0$$

The specification in equation 39 absorbs all variation at the destination–industry–year–type level, including $I_s x^k \lambda_{nt}$. It therefore cannot identify the destination-institution channel directly. We relax this in equation 40:

(40)

$$\log X^{kz}_{jnt} \approx \zeta^k_{jt} + \zeta^k_{nt} + \zeta^{kz}_t + \zeta_{jnt} + I_s\Big[\beta_\mu \mu_{jt} + \beta_\lambda \lambda_{nt} + \beta_{\lambda x}\left(x^k \times \lambda_{nt}\right) + \beta_{\mu\lambda}\left(\mu_{jt} \times \lambda_{nt}\right)\Big] + \varepsilon^{kz}_{jnt}$$

Here, we added ζ^k_{nt} are destination–industry–year fixed effects, and ζ^{kz}_t are industry–year–type fixed effects. These fixed effects still absorb a rich set of alternative determinants of trade flows, while allowing us to test for the role of destination institutions. The first-order approximation implies:

$$\beta_\lambda \equiv -\theta\ell_\lambda \geq 0, \qquad \beta_{\lambda x} = -\theta\ell_{\lambda x} \geq 0$$

The empirical strategy is thus disciplined by the model in two ways. First, all institutional variables are interacted with the specific-input indicator I_s, because the theoretical wedge applies only to specific inputs. Final goods and generic inputs serve as comparison groups that absorb broader institutional channels unrelated to the hold-up mechanism. Second, the key regressor is not simply the level of origin institutions, but also the destination institutions and their interaction.

6.1.2. Results

The results are consistent with two of the three main predictions. First, they confirm that the quality of contract enforcement institutions positively affects the trade of specific inputs. The first row in Table 4 supports this using the origin's Rule of Law (RoL) index, while the second row does the same using the destination's.[22] The positive effect of the destination's institutions is consistent with our theory, but it is at odds with the results in Berkowitz, Moenius, and Pistor (2006).

Second, the theory predicts that the contract enforcement institutions of the trade partners are substitutes. This is supported by the third row, which shows a negative and statistically significant coefficient associated with the interaction between the two indices (for specific inputs).

Third, the model predicts that the contract enforcement institutions of the destination should be more important for specific inputs that are more intensive in know-how transfers, which is tested in rows (4)-(6). In Antràs (2003), this was measured with the R&D intensity (columns 2 and 4) and the skill intensity (columns 3 and 4) of the downstream sector. Since this is not observable in the data (I only see the industry of origin of the product), I use a weighted average of the downstream R&D intensities, calculated using our measure of R&D intensity by sector and information from input-output tables. In addition to using the resulting "average downstream measures" of R&D and skill intensity individually, I also included them jointly in column (4). Finally, column (5) uses the first principal component of the two. The results do not support this prediction since, in all cases, the coefficient has the opposite sign.

6.2. Production cost channel

6.2.1. Empirical strategy

To test for the production cost channel, we need to construct a sector-level measure reflecting how contracting frictions affect unit costs c_{nt}^k. First, we can use equations 29, 31, 33 and 34 to calculate the linear approximation $\log c_{nt}^k$, interpreting c_{nt}^k as a function of the set of $\{\Lambda_{jnt}^l\}_{j,l}$ around a point with no contracting frictions. This approximation can be simplified as a function of a country–industry–year fixed effect ζ_{nt}^k and an expression that depends on the contracting frictions of all countries that sell inputs.

$$\log c_{nt}^k \approx \zeta_{nt}^k + \sum_{l=1}^{K} \left(\gamma^{lk} \alpha^l\right) \left[\sum_{j=1}^{J} \pi_{jnt}^{ls} \ell_{jn}^l\right]$$

The summation term tells us that the unit cost of production of k-industry goods in country n depends on the weighted average of the contracting frictions affecting the sourcing of its inputs ℓ_{jn}^l. Since firms use inputs from different sectors, the first component of the weight represents the importance that the upstream sector l has in the production of goods in the

[22] The direct effect of the quality of the destination's contract enforcement institutions is absorbed by the destination–industry–year–type fixed effect in column (1).

TABLE 4. Evidence for the trade cost channel.

Dependent variable: Bilateral exports by sector and type	(1)	(2)	(3)	(4)	(5)
(1) RS × Exporter RoL	4.179***	2.836***	2.874***	2.853***	2.846***
	(0.446)	(0.458)	(0.475)	(0.464)	(0.464)
(2) RS × Importer RoL		2.063***	2.674***	2.538***	1.867***
		(0.615)	(0.838)	(0.775)	(0.536)
(3) RS × Exporter RoL × Importer RoL	-5.609***	-3.743***	-3.817***	-3.780***	-3.765***
	(0.631)	(0.733)	(0.757)	(0.741)	(0.742)
RS × Importer RoL ×⋯					
(4) Avg. downstream R&D intensity		-3.829		-2.021	
		(2.340)		(2.179)	
(5) Avg. downstream skill intensity			-6.694*	-4.613	
			(3.515)	(3.022)	
(6) Avg. downstream R&D and skill intensity (PC)					-0.219*
					(0.123)
(7) Constant	16.623***	16.252***	16.249***	16.253***	16.254***
	(0.028)	(0.071)	(0.073)	(0.071)	(0.071)
Exporter-sector-year FE	X	X	X	X	X
Importer-sector-type-year FE	X				
Countries-year FE	X	X	X	X	X
Importer-sector-year FE		X	X	X	X
Sector-type-year FE		X	X	X	X
Number of observations	1,485,859	1,487,006	1,487,006	1,487,006	1,487,006

Clustered standard errors by country pair. *** p<0.01, ** p<0.05, * p<0.1. **RS** is a dummy variable that indicates whether trade flow corresponds to intermediate goods that require relationship-specific investments; **RoL** is the 'Rule of Law' index, our measure of the quality of contract enforcement institutions; **Downstream R&D** and **Downstream skill** are the average across downstream industries of R&D intensity and skill intensity, respectively; and **Downstream PC** is the predicted first principal component of Downstream R&D and skill intensities. All regressions were estimated using the module for Poisson pseudo-maximum-likelihood with multiple levels of fixed effects `ppmlhdfe` of Correia, Guimarães, and Zylkin (2020) in Stata SE, version 18.0.

downstream sector k (γ^{lk}). Given that not all inputs from each upstream sector are specific and only specific inputs are affected by contracting frictions, the second component of the weight represents the likelihood that an input from sector l is specific (α^l). Finally, since inputs can be purchased from different countries, the third component of the weight is the probability that a specific input from upstream industry l was purchased from the country j ($\pi^{l,s}_{jnt}$).

This term is closely related to the *contract intensity* measure in Nunn (2007). In fact, Nunn's measure is a special case in which all inputs are sourced domestically and cross-industry variation is ignored.[23] Under these assumptions, $\log c^k_{nt}$ can be written as a

[23]This implicit assumption is also present in the empirical specification of Levchenko (2007), although in his model the negotiation is not between firms but between capital and labor.

coefficient times the product of contract intensity and the quality of domestic institutions.

$$\log c_{nt}^{k,\text{Nunn}} \approx \zeta_{nt}^{k} - \beta \underbrace{\left[\sum_{l=1}^{K} \gamma^{lk} \alpha^{l}\right]}_{\substack{\text{Nunn's contract}\\ \text{intensity}}} \mu_{nt}$$

where γ^{lk} and α^{l} correspond to θ_{ij} and R_{j}^{neither} in Nunn (2007), respectively.[24]

Once we allow for international trade in inputs and for cross-sector variation in the importance of know-how transfers, we find that production costs are affected by contract-enforcement institutions in three ways. The first two highlight the role of domestic institutions via two modified "contract intensity" measures that account for (1) the share of domestic in expenditure of specific inputs $\pi_{nn}^{l,s}$ and (2) the importance of know-how transfers for the quality of inputs x^{k}. The third component shows how domestic costs are also affected by foreign institutions when specific inputs are imported.

$$\log c_{nt}^{k} \approx \zeta_{nt}^{k} + \overbrace{\underbrace{\beta_{\mu}\left[\sum_{l=1}^{K} \gamma^{lk} \alpha^{l} \pi_{nnt}^{l,s}\right]\mu_{nt}}_{\text{country } n \text{ as supplier}} + \underbrace{\beta_{\lambda}\left[\sum_{l=1}^{K} \gamma^{lk} \alpha^{l} x^{l}\right]\lambda_{nt}}_{\text{country } n \text{ as buyer}}}^{\text{domestic institutions}} + \underbrace{\beta_{\mu}\left\{\sum_{j \neq n}\left[\sum_{l=1}^{K} \gamma^{lk} \alpha^{l} \pi_{jnt}^{l,s}\right]\mu_{jt}\right\}}_{\text{foreign institutions}} \tag{41}$$

Testing for the production cost channel does not allow us to use origin–industry–year fixed effects since there would be no variation left to identify equation 41. Instead, we add three origin-industry interactions that account for the role of country endowments as a source of comparative advantage, as in Romalis (2004), Levchenko (2007) and Nunn (2007).

$$\begin{aligned} \log X_{nit}^{kz} = {} & \zeta_{it}^{kz} + \zeta_{nit} + b_{H}\left(\text{skill}^{k} \cdot H_{nt}\right) + b_{K}\left(\text{cap}^{k} \cdot K_{nt}\right) + b_{\mu}\Big(\sum_{l} \gamma^{lk} \alpha^{l} \pi_{nnt}^{l,s}\Big)\mu_{nt} \\ & + b_{\lambda}\Big[\sum_{l} \gamma^{lk} \alpha^{l} x^{l}\Big]\lambda_{nt} + b_{\mu}\Big(\sum_{j \neq n} \sum_{l} \gamma^{lk} \alpha^{l} \pi_{jnt}^{l,s} \mu_{jt}\Big) + \varepsilon_{nit}^{kz} \end{aligned} \tag{42}$$

where H_{nt} and K_{nt} are measures of origin's endowment of human and physical capital, and skill^{k} and cap^{k} are the skill and capital intensities in industry k. The first-order approximation implies:

$$b_{H} \geq 0, \qquad b_{K} \geq 0, \qquad b_{\mu} \geq 0, \qquad \text{and} \qquad b_{\lambda} \geq 0$$

[24] They are not exactly the same: α^{l} is equal to the share of specific inputs in sector k *output*, while R_{j}^{neither} is simply the proportion of inputs in sector k that are specific, that is, a *count* proportion.

6.2.2. Results

The results in Table 5 are consistent with the main predictions. First, they confirm that the effect of the quality of domestic contract enforcement institutions on sales is stronger in countries and industries that rely more on domestic suppliers (row 3).

Second, they show that the findings in Nunn (2007) reflect the mechanisms of this paper. When the interaction between Nunn's contract intensity measure and the exporter Rule of Law index is the only regressor accounting for contracting frictions, his results are replicated (column 1, row 4). However, once we include our measure for exposure to domestic and foreign institutions, the estimates change sign (columns 2-6) and in some cases are not statistically significant (column 2), in favor of the measure that accounts for the importance of domestic inputs in production.

However, when Nunn's regressor is interacted with the downstream average R&D share (row 5), skill intensity (row 6) or their principal component (row 7), the effect of contract intensity and domestic institutions becomes positive and statistically significant (columns 2 through 6). This suggests that for industries in which know-how transfers are important, domestic institutions have a second effect on contract-intensive industries beyond the one mediated by the share of domestic suppliers, as the theory predicts. Finally, while a higher exposure to foreign suppliers from countries with better institutions is positively associated with exports, these results are not statistically significant.

6.3. Both channels

6.3.1. Empirical strategy

Finally, we can modify the equations 39 and 40 to accommodate the equation 42 to jointly test for both channels. The augmented version of equation 39 is as follows.

$$\begin{aligned}\log X_{nit}^{k,z} =& \zeta_{it}^{kz} + \zeta_{nit} + b_H\left(\text{skill}^k \cdot H_{nt}\right) + b_K\left(\text{cap}^k \cdot K_{nt}\right) + b_\mu\Big(\sum_l \gamma^{lk}\alpha^l\pi_{nnt}^{l,s}\Big)\mu_{nt} \\ &+ b_\lambda\Big[\sum_l \gamma^{lk}\alpha^l x^l\Big]\lambda_{nt} + b_\mu\Big(\sum_{j\neq n}\sum_l \gamma^{lk}\alpha^l\pi_{jnt}^{l,s}\mu_{jt}\Big) \\ &+ \beta_\mu\Big(\mu_{nt}\times I_s\Big) + \beta_{\mu\lambda}\Big(\mu_{nt}\times\lambda_{it}\times I_s\Big) + \varepsilon_{nit}^{kz} \qquad (43)\end{aligned}$$

where we expect all coefficients to be positive except for $\beta_{\mu\lambda}$, which should be negative. In this regression, there are three coefficients associated with *domestic institutions*. Coefficients b_μ and b_λ reflect the effect of domestic institutions on *production costs* through domestic sourcing of specific inputs and via the importance of know-how transfers to suppliers, respectively. Coefficient β_μ reflects the effect of domestic institutions on *trade itself* since it only activates when the good being sold is a specific input.

TABLE 5. Evidence for the production cost channel.

Dependent variable: Bilateral exports by sector and type	(1)	(2)	(3)	(4)	(5)	(6)
(1) Skill intens. × Human capital	-2.051*	-1.627*	-2.017**	-2.616***	-1.812*	-2.686***
	(1.125)	(0.950)	(0.883)	(0.913)	(0.930)	(0.870)
(2) Capital intens. × Capital per worker	-0.001	0.219	0.355	0.271	0.354	0.327
	(0.712)	(0.696)	(0.720)	(0.707)	(0.719)	(0.717)
(3) Exp. Dom. Inst. (suppliers) × Exporter RoL		16.089***	17.422***	15.140***	17.725***	15.902***
		(4.557)	(4.312)	(4.476)	(4.234)	(4.394)
(4) Exp. Foreign Institutions		2.388	2.153	0.824	2.481	0.910
		(4.499)	(4.360)	(4.364)	(4.280)	(4.351)
(5) Contract intens. × Exporter RoL	5.109***	-6.522	-15.568***	-12.081***	-14.948***	-7.852*
	(0.620)	(4.352)	(4.163)	(4.543)	(4.349)	(4.252)
(6) Contract intens. × Exporter RoL × R&D intens.			85.671***		92.471***	
			(7.216)		(9.363)	
(7) Contract intens. × Exporter RoL × Skill intens.				53.039***	-12.504	
				(7.589)	(8.878)	
(8) Contract intens. × Exporter RoL × PC						3.159***
						(0.295)
(9) Constant	16.760***	16.299***	16.460***	16.590***	16.403***	16.639***
	(0.790)	(0.745)	(0.736)	(0.744)	(0.746)	(0.738)
Importer-sector-type-year FE	X	X	X	X	X	X
Country-pair-year FE	X	X	X	X	X	X
Number of observations	1,089,323	1,089,323	1,089,323	1,089,323	1,089,323	1,089,323

Clustered standard errors by country pair. *** p<0.01, ** p<0.05, * p<0.1. **RoL** is the 'Rule of Law' index, our measure of the quality of contract enforcement institutions; and **PC** is the predicted first principal component of R&D and skill intensities. All regressions were estimated using the module for Poisson pseudo-maximum-likelihood with multiple levels of fixed effects `ppmlhdfe` of Correia, Guimarães, and Zylkin (2020) in Stata SE, version 18.0.

The augmented version of equation 40 is as follows.

$$
\begin{aligned}
\log X_{nit}^{kz} =& \zeta_{it}^{k} + \zeta_{t}^{kz} + \zeta_{nit} + b_H\big(\text{skill}^k \cdot H_{nt}\big) + b_K\big(\text{cap}^k \cdot K_{nt}\big) + b_\mu\Big(\sum_l \gamma^{lk}\alpha^l \pi_{nnt}^{l,s}\Big)\mu_{nt} \\
&+ b_\lambda\Big[\sum_l \gamma^{lk}\alpha^l x^l\Big]\lambda_{nt} + b_\mu\Big(\sum_{j\neq n}\sum_l \gamma^{lk}\alpha^l \pi_{jnt}^{l,s}\mu_{jt}\Big) \\
&+ \beta_\mu\Big(\mu_{nt} \times I_s\Big) + \beta_\lambda\Big(\lambda_{it} \times I_s\Big) + \beta_{\lambda x}\Big(\lambda_{it} \times x^k \times I_s\Big) \\
&+ \beta_{\mu\lambda}\Big(\mu_{nt} \times \lambda_{it} \times I_s\Big) + \varepsilon_{nit}^{kz}
\end{aligned} \tag{44}
$$

where we expect all coefficients to be positive except for $\beta_{\mu\lambda}$ which should be negative.

6.3.2. Results

When jointly tested, our previous results remain supported by the evidence. In particular, the trade cost (rows 9 and 10) and the production cost via domestic suppliers (row 3) are both significant and positive. This gives credibility to the idea that these are two distinct channels through which institutions affect trade. R&D and contract intensive sectors are, in addition to the previous two channels, affected by contracting frictions (row 6). Finally, the effect of foreign institutions remains not significant.

TABLE 6. Joint evidence for both channels.

	Dependent variable: Bilateral exports by sector and type	(1)	(2)	(3)	(4)	(5)	(6)
(1)	Skill intens. × Human capital	-2.333**	-2.778***	-3.342***	-2.552***	-3.440***	-3.440***
		(0.918)	(0.852)	(0.882)	(0.895)	(0.843)	(0.843)
(2)	Capital intens. × Capital per worker	0.162	0.272	0.201	0.271	0.245	0.245
		(0.704)	(0.730)	(0.715)	(0.730)	(0.726)	(0.726)
(3)	Exp. Dom. Insts. (suppliers) × Exporter RoL	19.787***	21.238***	18.739***	21.597***	19.586***	19.586***
		(4.324)	(4.022)	(4.216)	(3.946)	(4.108)	(4.108)
(4)	Exp. Foreign Institutions	5.881	5.779	4.218	6.168	4.395	4.395
		(4.177)	(3.964)	(4.004)	(3.886)	(3.959)	(3.959)
(5)	Contract intens. × Exporter RoL	-10.100**	-18.960***	-15.354***	-18.275***	-11.197***	
		(3.981)	(3.777)	(4.140)	(3.954)	(3.843)	
(6)	Contract intens. × Exporter RoL × R&D intens.		86.106***		94.034***		
			(7.012)		(9.142)		
(7)	Contract intens. × Exporter RoL × Skill intens.			52.540***	-14.606*		
				(7.451)	(8.714)		
(8)	Contract intens. × Exporter RoL × PC					3.161***	3.161***
						(0.289)	(0.289)
(9)	RS × Exporter RoL	3.228***	3.228***	3.226***	3.247***	3.221***	3.221***
		(0.575)	(0.577)	(0.583)	(0.579)	(0.582)	(0.582)
(10)	RS × Importer RoL	0.816	1.185*	1.808**	1.586**	0.938	0.938
		(0.540)	(0.668)	(0.875)	(0.807)	(0.577)	(0.577)
(11)	RS × Exporter RoL × Importer RoL	-3.197***	-3.184***	-3.232***	-3.203***	-3.207***	-3.207***
		(0.828)	(0.832)	(0.840)	(0.835)	(0.837)	(0.837)
RS × Importer RoL ×···							
(12)	Avg. downstream R&D intens.		-4.890**		-3.283		
			(2.476)		(2.305)		
(13)	Avg. downstream skill intens.			-7.472**	-4.069		
				(3.666)	(3.066)		
(14)	Avg. downstream R&D and skill intens. (PC)					-0.267**	-0.267**
						(0.131)	(0.131)
(15)	Constant	16.250***	16.455***	16.574***	16.392***	16.642***	16.642***
		(0.749)	(0.743)	(0.751)	(0.749)	(0.747)	(0.747)
Importer-sector-year FE		X	X	X	X	X	X
Sector-year-type FE		X	X	X	X	X	X
Countries-year FE		X	X	X	X	X	X
Number of observations		1,088,005	1,088,005	1,088,005	1,088,005	1,088,005	1,088,005

Clustered standard errors by country pair. *** p<0.01, ** p<0.05, * p<0.1. **RS** is a dummy variable that indicates whether trade flow corresponds to intermediate goods that require relationship-specific investments; **RoL** is the 'Rule of Law' index, our measure of the quality of contract enforcement institutions; and **PC** is the predicted first principal component of R&D and skill intensities. All regressions were estimated using the module for Poisson pseudo-maximum-likelihood with multiple levels of fixed effects `ppmlhdfe` of Correia, Guimarães, and Zylkin (2020) in Stata SE, version 18.0.

7. Conclusion

Summary. This paper revisits the relationship between contract enforcement and international trade. Existing work has shown that countries with stronger domestic institutions specialize in sectors that rely more heavily on relationship-specific investments. The argument in this paper is that this mechanism is incomplete in an open economy because firms can source specific inputs from abroad and because the relevant contracting relationship may involve parties located in different institutional environments. To formalize this idea, I extend a model of incomplete contracts to allow input suppliers to differ by country and to allow both suppliers and buyers to make relationship-specific investments. The resulting contracting distortion can be summarized by a bilateral, industry-specific wedge that enters trade in specific inputs like an iceberg trade cost.

The model delivers two main channels linking institutions and trade. The first is a *trade cost channel*: weak contract enforcement raises the effective cost of trading specific inputs. This channel is specific to relationship-specific inputs and depends on the institutions of both the exporting and importing country. The second is a *production cost channel*: weak contract enforcement raises the cost of producing goods that rely heavily on specific inputs. This channel affects downstream trade through the price index of specific inputs and therefore depends on the sourcing structure of production. The model also predicts that exporter and importer institutions are substitutes and that firms may partially mitigate weak domestic institutions by importing specific inputs from countries with stronger contract enforcement.

The empirical results provide support for the two cost channels. Trade in specific inputs is higher when contract enforcement is stronger in either the origin or the destination country, and the negative interaction between origin and destination institutions is consistent with institutional substitutability. Domestic contract enforcement also matters more for exports in countries and industries that rely more heavily on domestic suppliers of specific inputs, and it matters especially in contract-intensive industries where know-how transfers are important. By contrast, the direct evidence for the bypass channel is weaker. Although the motivating evidence suggests that access to foreign institutions helps explain specialization in contract-intensive sectors, the main bilateral estimates do not show a statistically significant effect of exposure to foreign suppliers with stronger institutions.

Implications. The results imply that the institutional foundations of comparative advantage should not be interpreted solely as a domestic production-cost mechanism. Domestic institutions remain important, but their effect depends on the organization of input sourcing. A country with weak contract enforcement may face a disadvantage in producing specific inputs, but downstream firms may be less constrained if they can source those inputs from countries with stronger institutions. Conversely, strong domestic institutions may matter most when specific inputs are non-tradable or when the buyer's own relationship-specific investments are important.

The paper also has implications for empirical work. Tests based on aggregate exports or country-industry specialization may conflate several mechanisms. A theoretically consistent empirical strategy should distinguish specific inputs from final goods and generic inputs; allow the institutions of both trade partners to matter; and separate the trade cost channel from the production cost channel. Bilateral product-level data are therefore essential for identifying how contracting frictions shape trade flows. This distinction helps clarify why importer institutions can matter for specific input trade even though much of the existing comparative-advantage literature emphasizes exporter institutions.

More broadly, the analysis suggests that the gains from better institutions propagate through input-output linkages and cross-border sourcing relationships, rather than only through final-goods production. Improvements in domestic contract enforcement can increase exports not only by lowering production costs for domestic firms, but also by making the country a more attractive supplier of specific inputs to foreign buyers. At the same time, access to foreign suppliers may reduce the dependence of downstream producers on domestic contracting institutions.

Limitations and next steps. The analysis has several limitations. First, the empirical specifications are based on first-order approximations to the model's nonlinear contracting wedge. This approach yields transparent estimating equations and clear tests of the model's main predictions, but it does not recover the structural parameters needed for counterfactual analysis. A natural next step is to estimate the model more directly and quantify how changes in contract enforcement affect trade flows, sourcing patterns, and welfare.

Second, the empirical measures are imperfect. Product classifications distinguish final goods, generic inputs, and specific inputs, but they cannot observe the actual contractual environment of each transaction. Similarly, R&D intensity and skill intensity are proxies for the importance of know-how transfers, rather than direct measures of buyer-side relationship-specific investment. Future work could improve on these measures using firm-level sourcing data, customs transactions linked to input-output relationships, or information on contracts and supplier-buyer relationships.

Third, the current framework treats sourcing opportunities and product classifications as given. In practice, firms may respond to weak institutions by changing product design, vertically integrating, building long-term relationships, or choosing suppliers with better reputations. Incorporating these margins would help connect the trade-flow evidence in this paper to the broader literature on firm boundaries and global value chains.

Finally, the weak empirical support for the bypass channel deserves further investigation. It may reflect measurement error in exposure to foreign institutions, limited tradability of the most relationship-specific inputs, or heterogeneity across firms in their ability to access foreign suppliers. Understanding when foreign sourcing can and cannot substitute for domestic institutions is an important direction for future research.

References

Acemoglu, Daron, Pol Antràs, and Elhanan Helpman. 2007. "Contracts and Technology Adoption." *American Economic Review* 97 (3): 916–943.

Acemoglu, Daron, and Simon Johnson. 2005. "Unbundling Institutions." *Journal of Political Economy* 113 (5): 949–995.

Anderson, James E., and Douglas Marcouiller. 2002. "Insecurity and the Pattern of Trade: An Empirical Investigation." *The Review of Economics and Statistics* 84 (2): 342–352.

Antràs, Pol. 2003. "Firms, Contracts, and Trade Structure*." *The Quarterly Journal of Economics* 118 (4): 1375–1418.

Antràs, Pol. 2005. "Incomplete Contracts and the Product Cycle." *American Economic Review* 95 (4): 1054–1073.

Antràs, Pol, and Davin Chor. 2013. "Organizing the Global Value Chain." *Econometrica* 81 (6): 2127–2204.

Antràs, Pol, Evgenii Fadeev, Teresa C. Fort, and Felix Tintelnot. 2024. "Exporting, Global Sourcing, and Multinational Activity: Theory and Evidence from the United States." *The Review of Economics and Statistics*: 1–48.

Antràs, Pol, Teresa C. Fort, and Felix Tintelnot. 2017. "The Margins of Global Sourcing: Theory and Evidence from U.S. Firms." *American Economic Review* 107 (9): 2514–64.

Antràs, Pol, and Elhanan Helpman. 2004. "Global Sourcing." *Journal of Political Economy* 112 (3): 552–580.

Antràs, Pol, and Elhanan Helpman. 2008. *Contractual Frictions and Global Sourcing*. 9–54: Harvard University Press.

Berkowitz, Daniel, Johannes Moenius, and Katharina Pistor. 2004. "Legal Institutions and International Trade Flows." *Michigan Journal of International Law* 26 (1): 163–198.

Berkowitz, Daniel, Johannes Moenius, and Katharina Pistor. 2006. "Trade, Law, and Product Complexity." *The Review of Economics and Statistics* 88 (2): 363–373.

Caliendo, Lorenzo, and Fernando Parro. 2015. "Estimates of the Trade and Welfare Effects of NAFTA." *The Review of Economic Studies* 82 (1): 1–44.

Che, Yeon-Koo, and Donald B. Hausch. 1999. "Cooperative Investments and the Value of Contracting." *American Economic Review* 89 (1): 125–147.

Correia, Sergio, Paulo Guimarães, and Tom Zylkin. 2020. "Fast Poisson estimation with high-dimensional fixed effects." *The Stata Journal: Promoting communications on statistics and Stata* 20 (1): 95–115.

Costinot, Arnaud. 2009. "On the origins of comparative advantage." *Journal of International Economics* 77 (2): 255–264.

Costinot, Arnaud, Dave Donaldson, and Ivana Komunjer. 2012. "What Goods Do Countries Trade? A Quantitative Exploration of Ricardo's Ideas." *The Review of Economic Studies* 79 (2): 581–608.

Eaton, Jonathan, and Samuel Kortum. 2002. "Technology, Geography, and Trade." *Econometrica* 70 (5): 1741–1779.

Gaulier, Guillaume, and Soledad Zignago. 2010. "BACI: International Trade Database at the Product-Level. The 1994-2007 Version." Working Papers 2010-23, CEPII.

Grossman, Gene M., and Elhanan Helpman. 2002. "Integration versus Outsourcing in Industry Equilibrium." *The Quarterly Journal of Economics* 117 (1): 85–120.

Grossman, Sanford J, and Oliver D Hart. 1986. "The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration." *Journal of Political Economy* 94 (4): 691–719.

Hart, Oliver, and Bengt Holmström. 1987. *The theory of contracts*. 71–156, Econometric Society Monographs: Cambridge University Press.

Hart, Oliver, and J. Moore. 1990. "Property Rights and the Nature of the Firm." *Journal of Political Economy* 98 (6). Reprinted in numerous volumes.

Johnson, Robert C., and Guillermo Noguera. 2012. "Accounting for intermediates: Production sharing and trade in value added." *Journal of International Economics* 86 (2): 224–236.

Kaufmann, Daniel, and Aart Kraay. 2022. "Worldwide Governance Indicators, 2023 Update." `http://www.govindicators.org/`.

Kaufmann, Daniel, Aart Kraay, and Massimo Mastruzzi. 2004. "Governance Matters III: Governance Indicators for 1996, 1998, 2000, and 2002." *The World Bank Economic Review* 18 (2): 253–287.

Levchenko, Andrei A. 2007. "Institutional Quality and International Trade." *The Review of Economic Studies* 74 (3): 791–819.

Maskin, Eric, and Jean Tirole. 1999. "Unforeseen Contingencies and Incomplete Contracts." *The Review of Economic Studies* 66 (1): 83–114.

Nunn, Nathan. 2007. "Relationship-Specificity, Incomplete Contracts, and the Pattern of Trade*." *The Quarterly Journal of Economics* 122 (2): 569–600.

Nunn, Nathan, and Daniel Trefler. 2014. "Chapter 5 - Domestic Institutions as a Source of Comparative Advantage." In *Handbook of International Economics*, edited by Gita Gopinath, Elhanan Helpman, and Kenneth Rogoff, vol. 4 of Handbook of International Economics, 263–315: Elsevier.

Rauch, James E. 1999. "Networks versus markets in international trade." *Journal of International Economics* 48 (1): 7–35.

Rogerson, William P. 1992. "Contractual Solutions to the Hold-Up Problem." *The Review of Economic Studies* 59 (4): 777–793.

Romalis, John. 2004. "Factor Proportions and the Structure of Commodity Trade." *American Economic Review* 94 (1): 67–97.

Schmidt-Eisenlohr, Tim. 2013. "Towards a theory of trade finance." *Journal of International Economics* 91 (1): 96–112.

Segal, Ilya. 1999. "Complexity and Renegotiation: A Foundation for Incomplete Contracts." *The Review of Economic Studies* 66 (1): 57–82.

Shapley, L. S. 1953. *17. A Value for n-Person Games*. 307–318. Princeton: Princeton University Press.

Tirole, Jean. 1999. "Incomplete Contracts: Where Do We Stand?" *Econometrica* 67 (4): 741–781.

Winter, Eyal. 2002. "Chapter 53 The Shapley value." In *Handbook of Game Theory with Economic*

Applications, vol. 3 of Handbook of Game Theory with Economic Applications, 2025–2054: Elsevier.

Appendix A. Proofs for Section 3 (A model of input sourcing under incomplete contracts)

PROPOSITION 1 (Rent sharing).

a. *The supplier of input* ν *receives the following delivery payment:*

$$s(\nu) = \left(\frac{\alpha}{\alpha+\beta}\right)\left[\frac{m(\nu)}{M}\right]^{\beta} R \tag{A17}$$

b. *The buyer retains the following revenue:*

$$s_b = \left(\frac{\beta}{\alpha+\beta}\right) R \tag{A18}$$

PROOF. Suppose first that the number of intermediate-input suppliers is finite and equal to $\mathcal{I}$. The unit interval of input varieties is partitioned into $\mathcal{I}$ subintervals, each of length $\varepsilon = 1/\mathcal{I}$, and each supplier controls one such subinterval. The Shapley value of supplier ν is its *average marginal contribution* to revenue across all possible coalitions and all possible orders in which players can join those coalitions.

Revenue with n suppliers. Using equations 3, 4 and 5, the revenue generated by a coalition that includes the buyer and n suppliers is

$$R(n) = A\left[\sum_{i=1}^{n} \varepsilon m_i^{\beta}\right]^{\frac{\alpha}{\beta}} \tag{A1}$$

where $A \equiv A_1 A_2 \alpha^{-\alpha}$.

Marginal contribution. Suppose supplier ν joins a coalition that already includes the buyer and $n-1$ other suppliers. Its marginal contribution to revenue is:

$$\Delta R_{\nu}(n,\varepsilon) = A\left\{\left[\sum_{i=1}^{n-1} \varepsilon m_i^{\beta} + \varepsilon m_{\nu}^{\beta}\right]^{\frac{\alpha}{\beta}} - \left[\sum_{i=1}^{n-1} \varepsilon m_i^{\beta}\right]^{\frac{\alpha}{\beta}}\right\}$$

Relevant coalitions. If the buyer is not part of a coalition, the coalition generates no revenue. In that case, the value of the coalition is simply the sum of the outside values of its members. Since outside values are zero, a supplier's marginal contribution is positive only when the buyer is already in the coalition.

The Shapley value. There are $\mathcal{J}$ suppliers and one buyer, denoted by player 0. Let G be the set of all $(\mathcal{J}+1)!$ permutations of these players. For a permutation $g \in G$, let z_g^ν denote the set of players that precede supplier ν in that permutation. The Shapley value of supplier ν is:

$$S_\nu = \frac{1}{(\mathcal{J}+1)!} \sum_{g \in G} \left[v\left(z_g^\nu \cup \nu\right) - v(z_g^\nu) \right]$$

It is useful to rewrite this expression in terms of the position of supplier ν. If supplier ν is placed in position n, then there are n players before it and $\mathcal{J} - n$ players after it. Holding this position fixed, there are $\mathcal{J}!$ possible orderings of the remaining players. The probability that the buyer is among the n players that precede supplier ν is $n/\mathcal{J}$. Therefore,

$$\begin{aligned} S_\nu &= \frac{1}{(\mathcal{J}+1)!} \sum_{n=1}^{\mathcal{J}} \mathcal{J}! \left(\frac{n}{\mathcal{J}}\right) \Delta R_\nu(n, \varepsilon) \\ &= \frac{1}{(\mathcal{J}+1)\mathcal{J}} \sum_{n=1}^{\mathcal{J}} n \Delta R_\nu(n, \varepsilon) \\ &= \left(\frac{1}{1+\varepsilon}\right) \sum_{n=1}^{1/\varepsilon} (n\varepsilon) \Delta R_\nu(n, \varepsilon) \varepsilon \end{aligned} \tag{A2}$$

where the last line uses $\mathcal{J} = 1/\varepsilon$.

Linear approximation. Following Acemoglu, Antràs, and Helpman (2007), we approximate the marginal contribution of supplier ν around $m_\nu^\beta = 0$, retaining the first-order term in m_ν^β. This gives:

$$\Delta R_\nu(n, \varepsilon) \approx \left(\frac{\alpha}{\beta}\right) A m_\nu^\beta \varepsilon \left[\sum_{i=1}^{n-1} \varepsilon m_i^\beta \right]^{\frac{\alpha}{\beta} - 1}$$

Substituting this expression into equation A2 yields:

$$S_\nu \approx \left(\frac{\alpha}{\beta}\right) A m_\nu^\beta \varepsilon \left(\frac{1}{1+\varepsilon}\right) \sum_{n=1}^{1/\varepsilon} (n\varepsilon) \left[\sum_{i=1}^{n-1} \varepsilon m_i^\beta \right]^{\frac{\alpha}{\beta} - 1} \varepsilon$$

Asymptotic Shapley value. Divide both sides by ε and take the limit as $\mathcal{J} \to \infty$, equivalently as $\varepsilon \to 0$. Let $y = n\varepsilon$ denote the measure of suppliers that precede supplier ν in the ordering. In the limit, the external summation over coalition positions converges to an integral over $y \in [0, 1]$. The internal summation captures the aggregate quality contribution of the suppliers that precede ν. Because the Shapley value averages over all possible orderings with equal weight, conditional on a mass y of suppliers preceding ν, the set of suppliers that precedes ν is an asymptotically representative subset of measure y. Hence, their aggregate quality contribution is:

$$\sum_{i=1}^{n-1} \varepsilon m_i^\beta = (n-1)\varepsilon \left[\frac{1}{n-1} \sum_{i=1}^{n-1} m_i^\beta \right] \to y \int_0^1 m(\sigma)^\beta d\sigma = yM^\beta$$

Therefore, the asymptotic Shapley value of supplier ν is

$$\begin{aligned} S(\nu) \equiv \lim_{\varepsilon \to 0} \left(\frac{S_\nu}{\varepsilon} \right) &= \left(\frac{\alpha}{\beta} \right) Am(\nu)^\beta \left\{ \int_0^1 y \left[yM^\beta \right]^{\frac{\alpha}{\beta}-1} dy \right\} \\ &= \left(\frac{\alpha}{\beta} \right) Am(\nu)^\beta M^{\alpha-\beta} \int_0^1 y^{\frac{\alpha}{\beta}} dy \\ &= \left(\frac{\alpha}{\beta} \right) Am(\nu)^\beta M^{\alpha-\beta} \left(\frac{\beta}{\alpha+\beta} \right) \\ &\underbrace{=}_{\text{eq. } A1} \left(\frac{\alpha}{\alpha+\beta} \right) \left[\frac{m(\nu)}{M} \right]^\beta \underbrace{AM^\alpha}_{R} \end{aligned}$$

The buyer's asymptotic Shapley value is the residual revenue after paying all suppliers:

$$\begin{aligned} S_b &= R - \int_0^1 S(\nu) d\nu \\ &= R - \left(\frac{\alpha}{\alpha+\beta} \right) R \int_0^1 \left[\frac{m(\nu)}{M} \right]^\beta d\nu \\ &= R - \left(\frac{\alpha}{\alpha+\beta} \right) R \\ &= \left(\frac{\beta}{\alpha+\beta} \right) R \end{aligned}$$

where the third line uses $M^\beta = \int_0^1 m(\nu)^\beta d\nu$. □

PROPOSITION 2 (Tasks, services, and input quality).

Let $\psi \equiv x\lambda + (1-x)\mu$ *denote the effective degree of contractibility of input quality, and let*

$$\sigma_b \equiv \frac{s_b}{R} = \frac{\beta}{\alpha+\beta}$$

denote the buyer's share of revenue under the Shapley allocation. Then the incomplete-contracting equilibrium is characterized as follows.

a. *Nonverifiable tasks:*

$$\frac{a_n(\nu)}{a^{CC}(\nu)} = \frac{h_n(\nu)}{h^{CC}(\nu)} = \left\{ \sigma_b^{1-\beta\psi} \left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta} \right]^{\beta\psi} \right\}^{\frac{1}{1-\beta}} \equiv (\Gamma_n)^{\frac{1}{1-\beta}} \tag{A19}$$

b. *Verifiable tasks:*

$$\text{(A20)} \qquad \frac{a_c(\nu)}{a^{CC}(\nu)} = \frac{h_c(\nu)}{h^{CC}(\nu)} = \left\{ \sigma_b^{\beta(1-\psi)} \left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta} \right]^{1-\beta(1-\psi)} \right\}^{\frac{1}{1-\beta}} \equiv (\Gamma_c)^{\frac{1}{1-\beta}}$$

c. *Assembly services and know-how transfer:*

$$\text{(A21)} \qquad \frac{a(\nu)}{a^{CC}(\nu)} = \left\{ \sigma_b^{1-(1-\beta)\mu-\beta\psi} \left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta} \right]^{(1-\beta)\mu+\beta\psi} \right\}^{\frac{1}{1-\beta}} = \left(\Gamma_n^{1-\mu} \Gamma_c^{\mu} \right)^{\frac{1}{1-\beta}}$$

$$\text{(A22)} \qquad \frac{h(\nu)}{h^{CC}(\nu)} = \left\{ \sigma_b^{1-(1-\beta)\lambda-\beta\psi} \left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta} \right]^{(1-\beta)\lambda+\beta\psi} \right\}^{\frac{1}{1-\beta}} = \left(\Gamma_n^{1-\lambda} \Gamma_c^{\lambda} \right)^{\frac{1}{1-\beta}}$$

d. *Input quality:*

$$\text{(A23)} \qquad \frac{m(\nu)}{m^{CC}(\nu)} = \left\{ \sigma_b^{1-\psi} \left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta} \right]^{\psi} \right\}^{\frac{1}{1-\beta}} = \left(\Gamma_n^{1-\psi} \Gamma_c^{\psi} \right)^{\frac{1}{1-\beta}} \equiv (\Gamma)^{\frac{1}{1-\beta}}$$

PROOF. All agents anticipate the outcome of date-2 bargaining. Therefore, when they choose nonverifiable task levels at date 1, they take into account the payments implied by the Shapley allocation. At date 1, suppliers and the buyer choose nonverifiable task levels noncooperatively. The supplier of input ν chooses $a_n(\nu)$ to maximize its expected payoff from renegotiation net of the cost of nonverifiable assembly tasks:

$$\max_{a_n(\nu)} \left\{ s(\nu) - c_a(\nu)(1-\mu)a_n(\nu) \right\}$$

subject to equations 7 and 17. Similarly, the buyer chooses $h_n(\nu)$ to maximize its Shapley share of revenue net of the cost of nonverifiable know-how-transfer tasks:

$$\max_{\{h_n(\nu)\}} \left\{ s_b\Big(h_n(\nu)\Big) - \int_0^1 c_h(1-\lambda)h_n(\nu)d\nu \right\}$$

subject to equations 3–6 and 8. Because each supplier is infinitesimal, it takes aggregate objects such as M and R as given. The buyer, by contrast, internalizes how its choices affect aggregate revenue. The first-order conditions imply:

$$\text{(A3)} \quad a_n(\nu) = \frac{1-x}{c_a(\nu)} \left\{ \left[\left(\frac{a_c(\nu)}{1-x} \right)^{(1-x)\mu} \left(\frac{h_c(\nu)}{x} \right)^{x\lambda} \right]^{\beta} \frac{s_b \alpha M^{-\beta}}{\left[c_a(\nu)^{(1-x)(1-\mu)} c_h^{x(1-\lambda)} \right]^{\beta}} \right\}^{\frac{1}{1-\beta(1-\psi)}}$$

$$\text{(A4)}\quad h_n(\nu) = \frac{x}{c_h}\left\{\left[\left(\frac{a_c(\nu)}{1-x}\right)^{(1-x)\mu}\left(\frac{h_c(\nu)}{x}\right)^{x\lambda}\right]^{\beta}\frac{s_b\alpha M^{-\beta}}{\left[c_a(\nu)^{(1-x)(1-\mu)}c_h^{x(1-\lambda)}\right]^{\beta}}\right\}^{\frac{1}{1-\beta(1-\psi)}}$$

where $\psi \equiv x\lambda + (1-x)\mu$ is the effective degree of contractibility of input quality.

Anticipating these date-1 choices, the buyer chooses $a_c(\nu)$, $h_c(\nu)$, and $f(\nu)$ at date 0 to maximize its payoff subject to suppliers' participation constraints:

$$\begin{aligned}\max_{\{a_c(\nu),h_c(\nu),f(\nu)\}} \quad & s_b - \int_0^1\left[c_h\lambda h_c(\nu) + c_h(1-\lambda)h_n(\nu) + f(\nu)\right]d\nu\\ \text{s.t.} \quad & f(\nu) + s(\nu) - c_a(\nu)\left[\mu a_c(\nu) + (1-\mu)a_n(\nu)\right] \geq 0, \qquad \forall \nu \in [0,1]\end{aligned}$$

Because the buyer has all ex-ante bargaining power, it chooses the upfront transfer so that each supplier's participation constraint binds:

$$f(\nu) = -\left\{s(\nu) - c_a(\nu)\left[\mu a_c(\nu) + (1-\mu)a_n(\nu)\right]\right\}$$

When this expression is negative, $f(\nu)$ is a fee paid by the supplier rather than an upfront payment from the buyer. This is consistent with suppliers bidding for contracts, since they anticipate receiving a positive payment at date 2.

Substituting the binding participation constraint into the buyer's objective and using the fact that the Shapley allocation exhausts revenue, the date-0 problem becomes

$$\begin{aligned}\max_{\{a_c(\nu),h_c(\nu)\}} \quad & R - \int_0^1\left\{c_a(\nu)\left[\mu a_c(\nu) + (1-\mu)a_n(\nu)\right] + c_h\left[\lambda h_c(\nu) + (1-\lambda)h_n(\nu)\right]\right\}d\nu\\ \text{s.t.} \quad & \text{equations A3 and A4}\end{aligned}$$

Thus, as in the complete-contracting benchmark, the buyer maximizes the net social value of the transaction. The difference is that the buyer must now respect the incentive constraints determining $a_n(\nu)$ and $h_n(\nu)$. Solving this constrained problem yields:

$$\text{(A5)}\qquad a_c(\nu) = \underbrace{\left(\frac{1-x}{c_a(\nu)}\right)\left[p(\nu)m^{CC}(\nu)\right]}_{a^{CC}(\nu)}\left\{\sigma_b^{\beta(1-\psi)}\left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta}\right]^{1-\beta(1-\psi)}\right\}^{\frac{1}{1-\beta}}$$

$$\text{(A6)}\qquad h_c(\nu) = \underbrace{\left(\frac{x}{c_h}\right)\left[p(\nu)m^{CC}(\nu)\right]}_{h^{CC}(\nu)}\left\{\sigma_b^{\beta(1-\psi)}\left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta}\right]^{1-\beta(1-\psi)}\right\}^{\frac{1}{1-\beta}}$$

Substituting equations A5 and A6 into equations A3 and A4, and simplifying, gives the equilibrium levels of nonverifiable tasks:

$$\text{(A7)} \qquad a_n(\nu) = \underbrace{\left(\frac{1-x}{c_a(\nu)}\right)\left[p(\nu)m^{CC}(\nu)\right]}_{a^{CC}(\nu)} \left\{\sigma_b^{1-\beta\psi}\left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta}\right]^{\beta\psi}\right\}^{\frac{1}{1-\beta}}$$

$$\text{(A8)} \qquad h_n(\nu) = \underbrace{\left(\frac{x}{c_h}\right)\left[p(\nu)m^{CC}(\nu)\right]}_{h^{CC}(\nu)} \left\{\sigma_b^{1-\beta\psi}\left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta}\right]^{\beta\psi}\right\}^{\frac{1}{1-\beta}}$$

Finally, using equations A5–A8, these task levels imply the following equilibrium levels of assembly services, know-how transfer, and input quality:

$$\text{(A9)} \qquad a(\nu) = \underbrace{\left(\frac{1-x}{c_a(\nu)}\right)\left[p(\nu)m^{CC}(\nu)\right]}_{a^{CC}(\nu)} \left\{\sigma_b^{1-(1-\beta)\mu-\beta\psi}\left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta}\right]^{(1-\beta)\mu+\beta\psi}\right\}^{\frac{1}{1-\beta}},$$

$$\text{(A10)} \qquad h(\nu) = \underbrace{\left(\frac{x}{c_h}\right)\left[p(\nu)m^{CC}(\nu)\right]}_{h^{CC}(\nu)} \left\{\sigma_b^{1-(1-\beta)\lambda-\beta\psi}\left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta}\right]^{(1-\beta)\lambda+\beta\psi}\right\}^{\frac{1}{1-\beta}},$$

$$\text{(A11)} \qquad m(\nu) = m^{CC}(\nu)\left\{\sigma_b^{1-\psi}\left[\frac{1-\sigma_b(1-\psi)\beta}{1-(1-\psi)\beta}\right]^{\psi}\right\}^{\frac{1}{1-\beta}}$$

These expressions establish the proportionality results in Proposition 2. □

PROPOSITION 3 (Effects of incomplete contracts on input quality).

a. *Incomplete contracts lead to underinvestment in all tasks. This distortion is larger for non-verifiable tasks than for verifiable tasks, resulting in lower input quality:*

$$0 \leq \Gamma_n \leq \Gamma_c \leq 1 \quad \Rightarrow \quad \Gamma \in [0,1]$$

b. *The input-quality distortion term* Γ *is increasing in contractibility, whether of buyer-provided services* (λ) *or supplier-provided services* (μ)*, and in the substitutability of specific inputs* (β)*. It is decreasing in the importance of specific inputs in production* (α)*:*

$$\Gamma(\overbrace{\alpha}^{-}, \overbrace{\beta}^{+}, \overbrace{\mu}^{+}, \overbrace{\lambda}^{+})$$

c. *Contracting distortions disappear if effective contractibility is complete,* $\psi \to 1$*. This happens* *if all tasks are fully verifiable,* $\mu \to 1$ *and* $\lambda \to 1$*, or if input quality depends only on the services provided by one party and those services are fully contractible:* $\lambda \to 1$ *and* $x \to 1$*, or* $\mu \to 1$ *and* $x \to 0$*. Contracting distortions also disappear if specific inputs are unimportant in production,* $\alpha \to 0$*. By contrast, input provision collapses as* $\beta \to 0$*:*

$$\lim_{\psi \nearrow 1} \Gamma = \lim_{\alpha \searrow 0} \Gamma = 1, \qquad \text{and} \qquad \lim_{\beta \searrow 0} \Gamma = 0$$

PROOF. Let:

$$B \equiv \frac{1 - \sigma_b(1-\psi)\beta}{1-(1-\psi)\beta}$$

The following standard bounds for logarithms will be useful:

$$1 - \frac{1}{x} \leq \log(x) \leq x - 1 \tag{A12}$$

Bounds. First, $\Gamma_n \geq 0$ because it is the product of $\sigma_b \in [0,1]$ and B, which is not only positive, but larger than one:

$$B = 1 + \frac{(1-\sigma_b)(1-\psi)\beta}{1-(1-\psi)\beta} \geq 1$$

Second, $\Gamma_c \leq 1$ because $\log \Gamma_c \leq 0$:

$$\begin{aligned}
\log \Gamma_c &= \beta(1-\psi)\log(\sigma_b) + [1-\beta(1-\psi)]\log(B) \\
&\leq \beta(1-\psi)(\sigma_b - 1) + [1-\beta(1-\psi)]\left[\frac{(1-\sigma_b)(1-\psi)\beta}{1-(1-\psi)\beta}\right] \\
&= 0
\end{aligned}$$

where we used the upper bound from equation A12. Finally, $\Gamma_n \leq \Gamma_c$ because $\sigma_b \leq 1 \leq B$:

$$\begin{aligned}
\Gamma_n \leq \Gamma_c &\Leftrightarrow \sigma_b^{1-\beta\psi} B^{\beta\psi} \leq \sigma_b^{\beta(1-\psi)} B^{1-\beta(1-\psi)} \\
&\Leftrightarrow \sigma_b^{1-\beta} \leq B^{1-\beta} \\
&\Leftrightarrow \sigma_b \leq B,
\end{aligned}$$

Since $\Gamma = \Gamma_n^{1-\psi}\Gamma_c^{\psi}$, the bounds $0 \leq \Gamma_n \leq \Gamma_c \leq 1$ imply $\Gamma \in [0,1]$.

Comparative statics of σ_b *and B.* Consistent with the intuition that the buyer's bargaining power is higher when inputs are easier to substitute and less important for production, the buyer's revenue share is increasing in input substitutability and decreasing in the importance of specific inputs in production:

$$\frac{\partial \sigma_b}{\partial \beta} = \frac{\sigma_b(1-\sigma_b)}{\beta} \geq 0, \qquad \frac{\partial \sigma_b}{\partial \alpha} = -\frac{\sigma_b(1-\sigma_b)}{\alpha} \leq 0$$

Next consider B. Differentiating with respect to β, taking into account the effect of β on σ_b, gives:

$$\frac{\partial B}{\partial \beta} = \frac{(1-\psi)(1-\sigma_b)\left[(1-\sigma_b)+\beta(1-\psi)\sigma_b\right]}{[1-\beta(1-\psi)]^2} \geq 0$$

Differentiating with respect to α gives:

$$\frac{\partial B}{\partial \alpha} = \frac{\sigma_b(1-\sigma_b)(1-\psi)\beta}{\alpha[1-\beta(1-\psi)]} \geq 0$$

Finally, differentiating with respect to ψ gives:

$$\frac{\partial B}{\partial \psi} = -\frac{\beta(1-\sigma_b)}{[1-\beta(1-\psi)]^2} \leq 0$$

Thus, B is increasing in β and α, but decreasing in effective contractibility ψ. Since $\psi = x\lambda + (1-x)\mu$, the effects of λ and μ operate through their effect on ψ.

Comparative statics of Γ. Since $\Gamma = \sigma_b^{1-\psi} B^{\psi}$, differentiating with respect to β and α yields:

$$\begin{aligned}
\frac{\partial \Gamma}{\partial \beta} &= \Gamma(1-\psi)(1-\sigma_b)\left\{\frac{\beta\psi + [1-\beta(1-\psi)](1-\beta\sigma_b)}{\beta[1-\beta(1-\psi)\sigma_b][1-\beta(1-\psi)]}\right\} \geq 0, \\
\frac{\partial \Gamma}{\partial \alpha} &= -\frac{\Gamma(1-\psi)\sigma_b(1-\sigma_b\beta)}{[1-\sigma_b(1-\psi)\beta]\beta} \leq 0
\end{aligned}$$

Differentiating with respect to ψ yields:

$$\frac{\partial \Gamma}{\partial \psi} = \Gamma\left\{\log\left[\frac{1-\sigma_b(1-\psi)\beta}{\sigma_b[1-(1-\psi)\beta]}\right] - \frac{\psi\beta(1-\sigma_b)}{[1-\sigma_b(1-\psi)\beta][1-(1-\psi)\beta]}\right\}$$

Using the lower bound in equation A12,

$$\frac{\partial \Gamma}{\partial \psi} \geq \frac{\Gamma(1-\sigma_b)(1-\beta)}{[1-\sigma_b(1-\psi)\beta][1-(1-\psi)\beta]} \geq 0$$

Limits. The limiting results follow directly from the expression for Γ. First, as specific inputs become less important in production, the buyer retains a larger share of revenue. This strengthens the buyer's investment incentives while weakening suppliers' incentives. In the limit, however, supplier incentives become irrelevant because specific inputs no longer affect production:

$$\lim_{\alpha \searrow 0} \sigma_b = 1 \quad \text{and} \quad \lim_{\alpha \searrow 0} B = 1 \quad \Rightarrow \quad \lim_{\alpha \searrow 0} \Gamma = 1$$

Second, as contractibility increases, the buyer gains control over the supplier's decisions, which increases input quality:

$$\lim_{\psi \nearrow 1} \sigma_b^{1-\psi} = 1 \quad \text{and} \quad \lim_{\psi \nearrow 1} B^{\psi} = 1 \quad \Rightarrow \quad \lim_{\psi \nearrow 1} \Gamma = 1$$

Finally, as specific inputs become more difficult to substitute, $\beta \searrow 0$, the elasticity of substitution across them converges to one and each supplier becomes essential. The buyer's revenue share therefore converges to zero and specific input sourcing collapses:

$$\lim_{\beta \searrow 0} \sigma_b = 0 \quad \text{and} \quad \lim_{\beta \searrow 0} B = 1 \quad \Rightarrow \quad \lim_{\beta \searrow 0} \Gamma = 0$$

These limits establish the final part of the proposition. □

PROPOSITION 4 (Effects of incomplete contracts on unit costs).

a. *Incomplete contracts raise the effective unit cost of input quality:*

$$\Lambda \in [1, +\infty)$$

b. *The effective unit cost of input quality decreases with contractibility, whether of buyer-provided tasks* (λ) *or supplier-provided tasks* (μ)*. It increases with the importance of specific inputs in production* (α)*:*

$$\Lambda(\overbrace{\alpha}^{+}, \overbrace{\mu}^{-}, \overbrace{\lambda}^{-})$$

If $\mu \leq \lambda$*, the effective unit cost of input quality also decreases with the elasticity of substitution across specific inputs:*

$$\mu \leq \lambda \Rightarrow \frac{d\Lambda}{d\beta} \leq 0$$

c. *Contracting unit cost distortions disappear if effective contractibility is complete,* $\psi \to 1$*. This happens if all tasks are fully verifiable,* $\mu \to 1$ *and* $\lambda \to 1$*, or if input quality depends only on assembly services and they are fully contractible,* $\mu \to 1$ *and* $x \to 0$*. Contracting unit cost distortions also disappear if specific inputs are unimportant in production,* $\alpha \to 0$*. By contrast, effective unit costs become arbitrarily large if specific inputs are essential,* $\beta \to 0$*:*

$$\lim_{\psi \nearrow 1} \Lambda = \lim_{\alpha \searrow 0} \Lambda = 1, \quad \textit{and} \quad \lim_{\beta \searrow 0} \Lambda = +\infty$$

PROOF. The following standard bounds for logarithms will be useful:

$$1 - \frac{1}{x} \leq \log(x) \leq x - 1 \tag{A13}$$

To economize on notation, define:

$$B \equiv \frac{1 - \sigma_b(1-\psi)\beta}{1-(1-\psi)\beta}, \qquad \Xi \equiv \left[\mu B + (1-\mu)\sigma_b\right]^{\frac{1-\beta}{\beta}}$$

Then equation 26 can be written as $\Lambda = (\Gamma\Xi)^{-1}$.

Lower bound of Λ. To show that $\Lambda \geq 1$, it is enough to show that $-\log(\Lambda) \leq 0$. Using $\Lambda = (\Gamma\Xi)^{-1}$:

$$\begin{aligned} -\log(\Lambda) &= \log(\Gamma) + \log(\Xi) \\ &= (1-\psi)\log(\sigma_b) + \psi\log(B) + \left(\frac{1-\beta}{\beta}\right)\log\left[\mu B + (1-\mu)\sigma_b\right] \end{aligned}$$

Applying the upper bound in equation A13,

$$\begin{aligned} -\log(\Lambda) &\leq (1-\psi)(\sigma_b - 1) + \psi(B-1) + \left(\frac{1-\beta}{\beta}\right)\left[\mu B + (1-\mu)\sigma_b - 1\right] \\ &= -\frac{(1-\sigma_b)(1-\beta)(1-\mu)}{\beta[1-\beta(1-\psi)]} \leq 0 \end{aligned}$$

Therefore, $\Lambda \geq 1$.

Comparative statics. Since $\Lambda = (\Gamma\Xi)^{-1}$, for any parameter z,

$$\frac{\partial\Lambda}{\partial z} = -\Lambda\left[\frac{1}{\Gamma}\left(\frac{\partial\Gamma}{\partial z}\right) + \frac{1}{\Xi}\left(\frac{\partial\Xi}{\partial z}\right)\right] = -\Lambda\left[\frac{\partial\log\Gamma}{\partial z} + \frac{\partial\log\Xi}{\partial z}\right]$$

First consider β. Proposition 3 established that Γ is increasing in β, so it remains to determine how Ξ changes. Differentiating Ξ with respect to β gives:

$$\frac{\partial\Xi}{\partial\beta} = \Xi\left\{\left(\frac{1-\beta}{\beta}\right)\frac{\mu\overbrace{\frac{\partial B}{\partial\beta}}^{\geq 0} + (1-\mu)\overbrace{\frac{\partial\sigma_b}{\partial\beta}}^{\geq 0}}{\mu B + (1-\mu)\sigma_b} - \frac{1}{\beta^2}\log\left[\mu B + (1-\mu)\sigma_b\right]\right\}$$

The first term inside braces is nonnegative because both $\partial B/\partial\beta$ and $\partial\sigma_b/\partial\beta$ are nonnegative. The logarithmic term is also nonnegative whenever $\mu B + (1-\mu)\sigma_b \leq 1$ since it enters with a negative sign. Using the definition of B, this condition is equivalent to $\mu \leq 1-\beta(1-\psi)$.

If $\mu \leq \lambda$, then $\mu \leq \psi$, and hence:

$$\mu \leq \psi \leq 1 - \beta + \beta\psi = 1 - \beta(1 - \psi)$$

Therefore, $\partial\Xi/\partial\beta \geq 0$ when $\mu \leq \lambda$. Since $\partial\Gamma/\partial\beta \geq 0$, it follows that:

$$\mu \leq \lambda \quad \Rightarrow \quad \frac{\partial\Lambda}{\partial\beta} \leq 0$$

Next consider α. In this case, focusing on Ξ alone is not sufficient. Since α affects Λ only through σ_b, differentiating with respect to α gives:

$$\frac{\partial\Lambda}{\partial\alpha} = -\Lambda\left\{\frac{(1-\psi)(1-\sigma_b\beta)}{\sigma_b\left[1-\sigma_b(1-\psi)\beta\right]} + \left(\frac{1-\beta}{\beta}\right)\left[\frac{1-\mu-(1-\psi)\beta}{\mu(1-\sigma_b)+\sigma_b\left[1-(1-\psi)\beta\right]}\right]\right\}\left(\underbrace{\frac{\partial\sigma_b}{\partial\alpha}}_{\leq 0}\right)$$

The first term inside braces is nonnegative. The sign of the second term is ambiguous, but, holding ψ fixed, that term is decreasing in μ. Therefore, the expression inside braces is bounded from below by its value at $\mu = 1$. Hence:

$$\begin{aligned}\frac{\partial\Lambda}{\partial\alpha} &\geq -\Lambda\left\{\frac{(1-\psi)(1-\sigma_b\beta)}{\sigma_b\left[1-\sigma_b(1-\psi)\beta\right]} - \left(\frac{1-\beta}{\beta}\right)\left[\frac{(1-\psi)\beta}{1-\sigma_b(1-\psi)\beta}\right]\right\}\left(\underbrace{\frac{\partial\sigma_b}{\partial\alpha}}_{\leq 0}\right) \\ &= -\left\{\frac{\Lambda(1-\psi)(1-\sigma_b)}{\sigma_b\left[1-\sigma_b(1-\psi)\beta\right]}\right\}\left(\underbrace{\frac{\partial\sigma_b}{\partial\alpha}}_{\leq 0}\right) \geq 0\end{aligned}$$

Since σ_b is decreasing in α, the effective unit cost of input quality is increasing in the importance of specific inputs in production.

Next consider supplier contractibility, μ. Proposition 3 established that Γ is increasing in μ, so it remains to determine how Ξ changes. Differentiating Ξ with respect to μ gives:

$$\begin{aligned}\frac{\partial\Xi}{\partial\mu} &= \Xi\left(\frac{1-\beta}{\beta}\right)\left[\frac{(B-\sigma_b)+\mu\frac{\partial B}{\partial\mu}}{\mu B+(1-\mu)\sigma_b}\right] \\ &= \Xi\left(\frac{1-\beta}{\beta}\right)\left\{\frac{1-\sigma_b}{\mu(1-\sigma_b)+\sigma_b\left[1-(1-\psi)\beta\right]}\right\}\left[\frac{1-(1-x\lambda)\beta}{1-(1-\psi)\beta}\right] \geq 0\end{aligned}$$

Therefore,

$$\frac{\partial\Lambda}{\partial\mu} = -\Lambda\left\{\underbrace{\frac{\partial\log\Gamma}{\partial\mu}}_{\geq 0} + \underbrace{\frac{\partial\log\Xi}{\partial\mu}}_{\geq 0}\right\} \leq 0$$

Thus, stronger supplier-side contractibility lowers the effective unit cost of input quality.

Next consider buyer contractibility, λ. In this case, focusing on Ξ alone is not sufficient because Ξ is decreasing in λ. Since $\partial\psi/\partial\lambda = x$, we have:

$$\frac{\partial\Lambda}{\partial\lambda} = -x\Lambda\left\{\underbrace{\log\left(\frac{B}{\sigma_b}\right)}_{\geq 0} + \left[\frac{\psi}{B} + \left(\frac{1-\beta}{\beta}\right)\left(\frac{\mu}{\mu B + (1-\mu)\sigma_b}\right)\right]\underbrace{\left(\frac{\partial B}{\partial\psi}\right)}_{\leq 0}\right\}$$

The first term inside braces is nonnegative because $B \geq \sigma_b$. The second term is nonpositive because $\partial B/\partial\psi \leq 0$. Holding ψ fixed, the term in square brackets is increasing in μ, so the expression inside braces is decreasing in μ. Since $\partial\Lambda/\partial\lambda$ is the negative of this expression times $x\Lambda$, it is bounded from above by its value at $\mu = 1$:

$$\begin{aligned}\frac{\partial\Lambda}{\partial\lambda} &\leq -x\Lambda\left\{\log\left(\frac{B}{\sigma_b}\right) + \left[\frac{\psi}{B} + \left(\frac{1-\beta}{\beta}\right)\left(\frac{1}{B}\right)\right]\left(\frac{\partial B}{\partial\psi}\right)\right\} \\ &= -x\Lambda\left[\log\left(\frac{B}{\sigma_b}\right) - \frac{1-\sigma_b}{1-\sigma_b(1-\psi)\beta}\right]\end{aligned}$$

Applying the lower bound in equation A13,

$$\begin{aligned}\frac{\partial\Lambda}{\partial\lambda} &\leq -x\Lambda\left[\left(1 - \frac{\sigma_b}{B}\right) - \frac{1-\sigma_b}{1-\sigma_b(1-\psi)\beta}\right] \\ &= -x\Lambda\left[\frac{1-\sigma_b}{1-\sigma_b(1-\psi)\beta} - \frac{1-\sigma_b}{1-\sigma_b(1-\psi)\beta}\right] \\ &= 0\end{aligned}$$

Thus, stronger buyer-side contractibility lowers the effective unit cost of input quality.

Limits. Proposition 3 already established the limits for Γ, so here we first show the limiting results for Ξ. First, as specific inputs become less important in production, the buyer retains a larger share of revenue. This strengthens the buyer's investment incentives while weakening suppliers' incentives. In the limit, however, supplier incentives become irrelevant because specific inputs no longer affect production:

$$\lim_{\alpha\searrow 0}\sigma_b = 1 \quad \text{and} \quad \lim_{\alpha\searrow 0} B = 1 \quad \Rightarrow \quad \lim_{\alpha\searrow 0}\Xi = 1$$

Second, as contractibility increases, the buyer gains control over the supplier's decisions, which increases input quality:[25]

$$\lim_{\substack{\psi \nearrow 1 \\ \mu \nearrow 1}} (1-\mu)\sigma_b = 0 \quad \text{and} \quad \lim_{\substack{\psi \nearrow 1 \\ \mu \nearrow 1}} \mu B = 1 \quad \Rightarrow \quad \lim_{\substack{\psi \nearrow 1 \\ \mu \nearrow 1}} \Xi = 1$$

Finally, as specific inputs become more difficult to substitute, $\beta \searrow 0$, the elasticity of substitution across them converges to one and each supplier becomes essential. The buyer's revenue share therefore converges to zero and specific input sourcing collapses:[26]

$$\lim_{\beta \searrow 0} \sigma_b = 0 \quad \text{and} \quad \lim_{\beta \searrow 0} B = 1 \quad \Rightarrow \quad \lim_{\beta \searrow 0} \left[\mu B + \sigma_b(1-\mu)\right] = \mu \quad \Rightarrow \quad \lim_{\beta \searrow 0} \Xi = \begin{cases} 0 & \text{if} \quad \mu \in [0,1) \\ e^{1-\psi} & \text{if} \quad \mu = 1 \end{cases}$$

The limiting results for Λ follow directly from the expression $\Lambda = (\Gamma\Xi)^{-1}$. If specific inputs become unimportant in production:

$$\lim_{\alpha \searrow 0} \Gamma = 1 \quad \text{and} \quad \lim_{\alpha \searrow 0} \Xi = 1 \quad \Rightarrow \quad \lim_{\alpha \searrow 0} \Lambda = 1$$

If effective contractibility becomes complete (with $x \neq 0$):

$$\lim_{\substack{\psi \nearrow 1 \\ \mu \nearrow 1}} \Gamma = 1 \quad \text{and} \quad \lim_{\substack{\psi \nearrow 1 \\ \mu \nearrow 1}} \Xi = 1 \quad \Rightarrow \quad \lim_{\substack{\psi \nearrow 1 \\ \mu \nearrow 1}} \Lambda = 1$$

Finally, as specific inputs become more difficult to substitute:

$$\lim_{\beta \searrow 0} \Gamma = 0 \quad \text{and} \quad \lim_{\beta \searrow 0} \Xi = \begin{cases} 0 & \text{if} \quad \mu \in [0,1) \\ e^{1-\psi} & \text{if} \quad \mu = 1 \end{cases} \quad \Rightarrow \quad \lim_{\beta \searrow 0} \Lambda = +\infty$$

These limits establish the final part of the proposition. □

[25] We do not consider the knife-edge case in which only know-how transfers matter, $\lambda \to 1$ and $x \to 1$, because in that case there would be no observed trade flows, $X^{IC}(\nu) \to 0$.

[26] The last equality considers two cases. When $\mu < 1$, $\Xi \to 0$ because a number below one is raised to a power that gets arbitrarily large. When $\mu = 1$, $\Xi \to e^{1-\psi}$ comes from applying l'Hôpital's rule.